AIML 231— Week 7

Clustering

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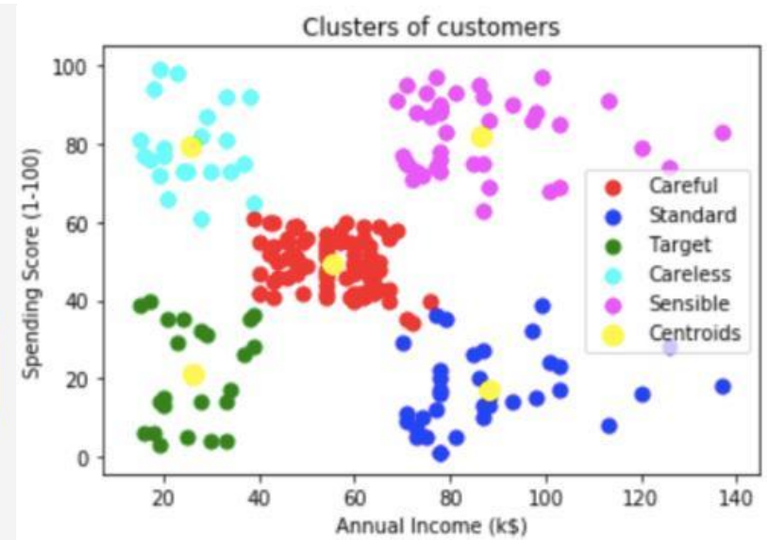
Lecture Overview

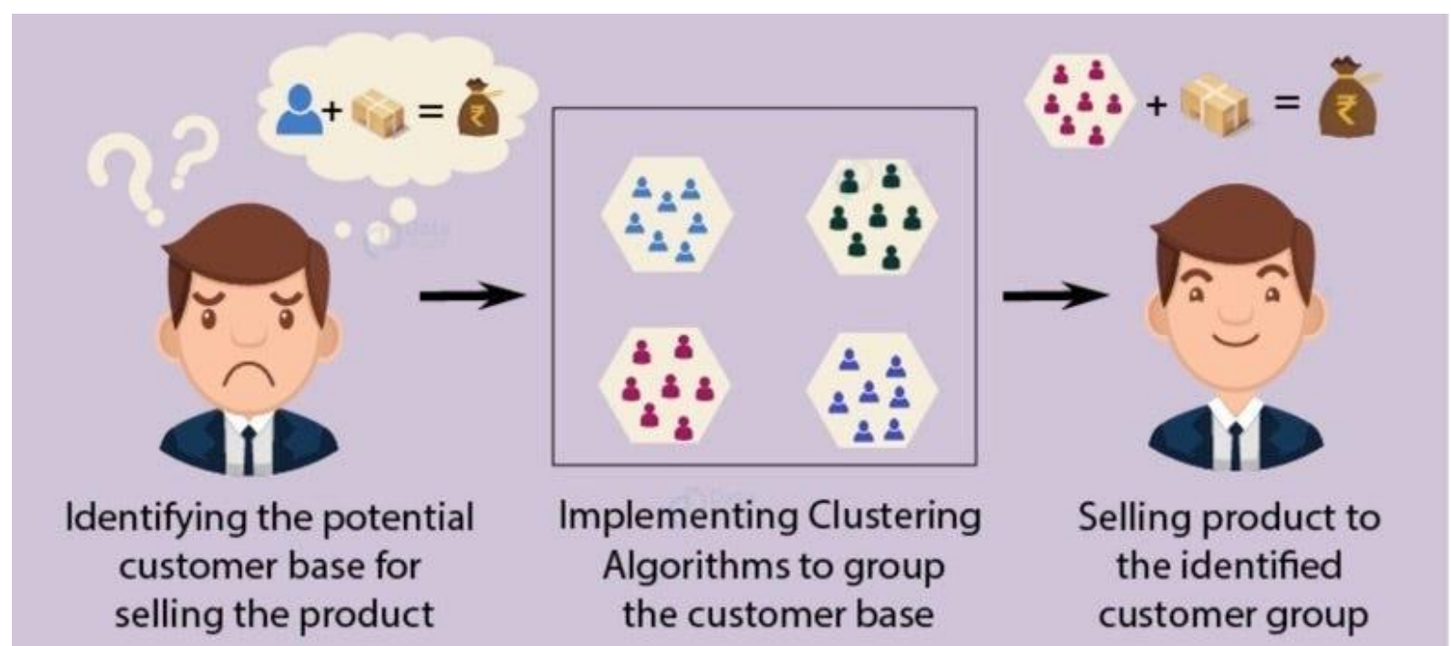
- ★ What is clustering?
- ★ Distance measures
- ★ Clustering Models: K-Means, Agglomerative Clustering
- ★ Clustering metric
- ★ Density Based Clustering

Customer Segmentation

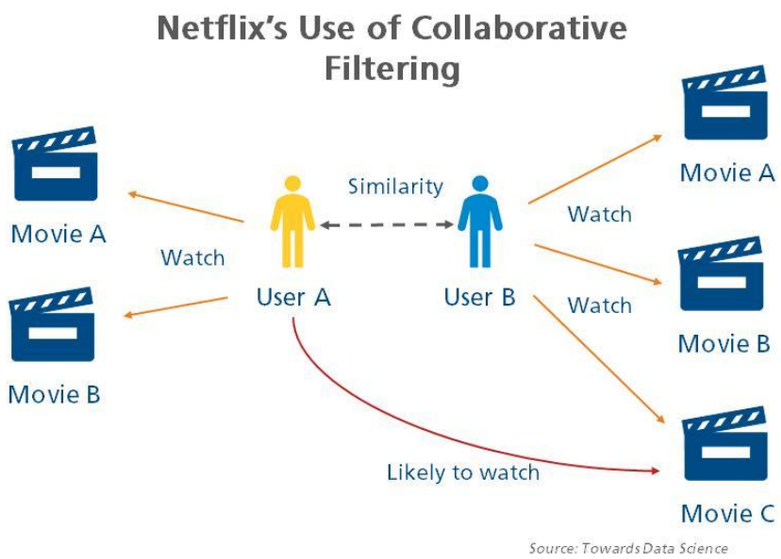
- Divide customers into similar groups based on common characteristics:
  - Historical purchases
  - Geographical locations
  - Products and services
  - Socio-economic: income, education

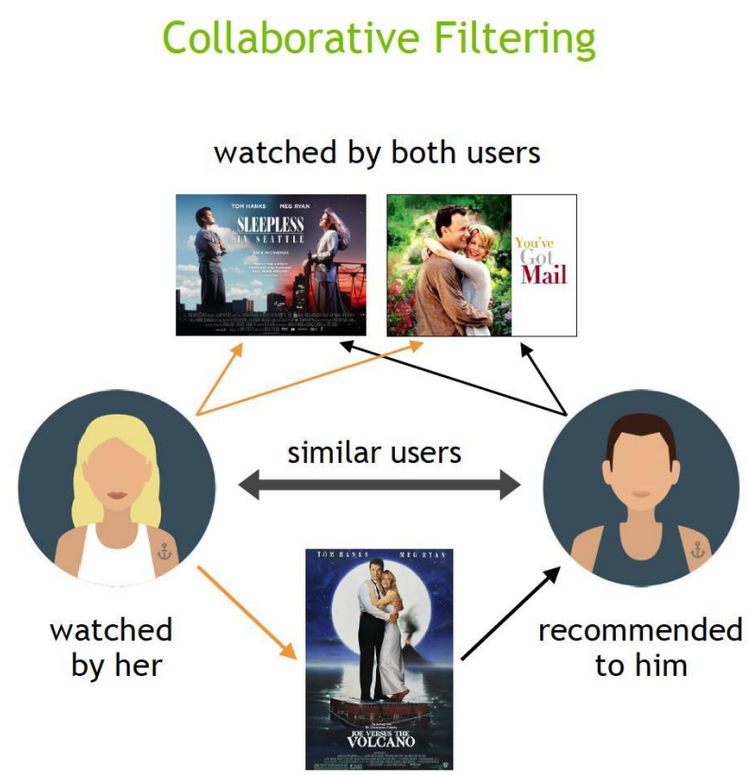


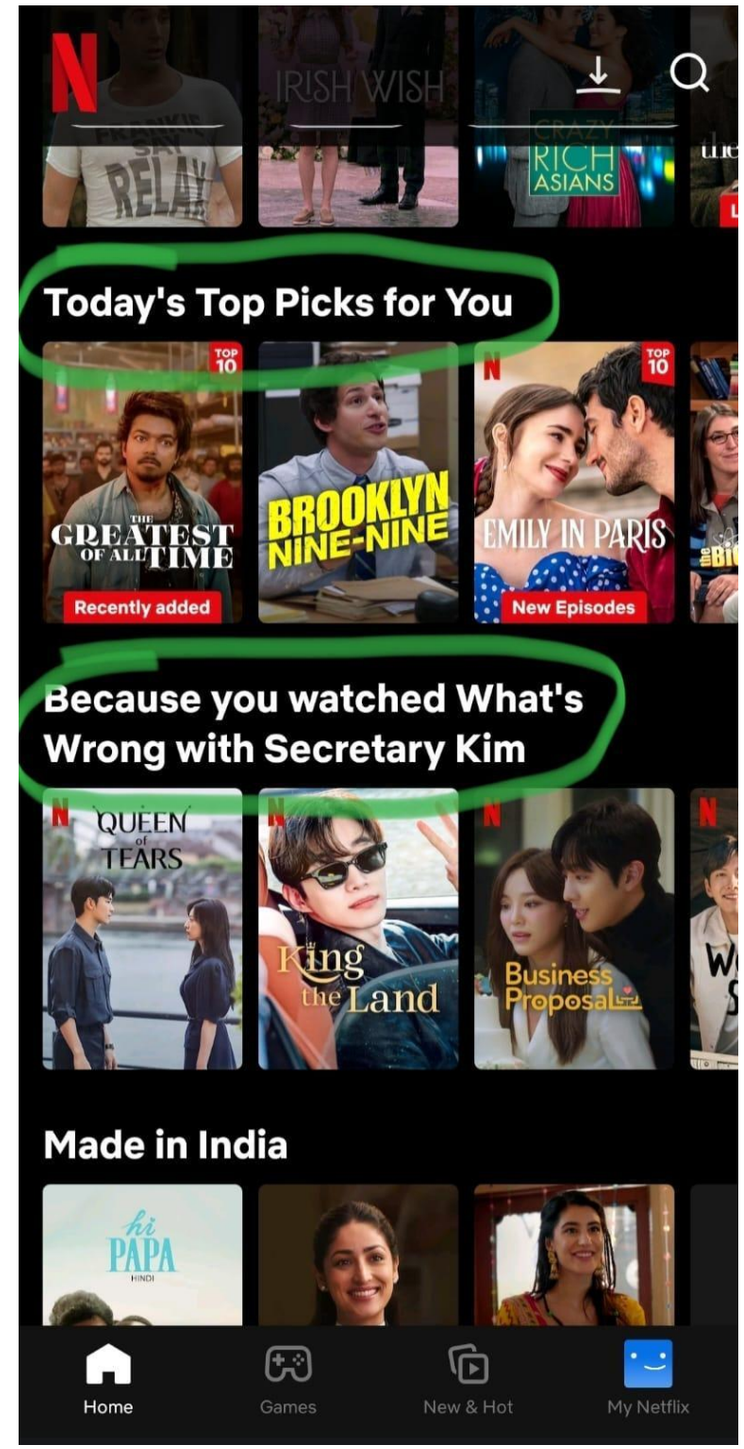
- Each group has its own effective marketing strategies
- Effective communication
- Better customer supports
- Increase revenue



Recommendation System in Netflix



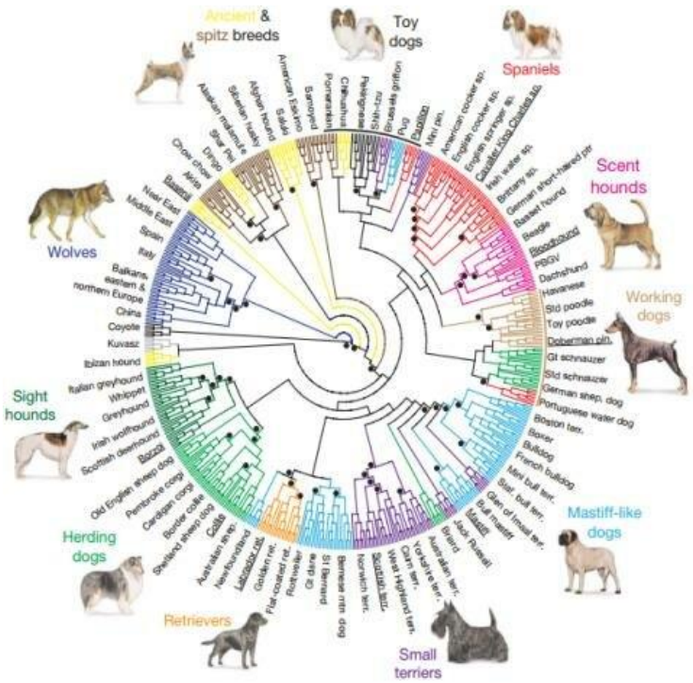


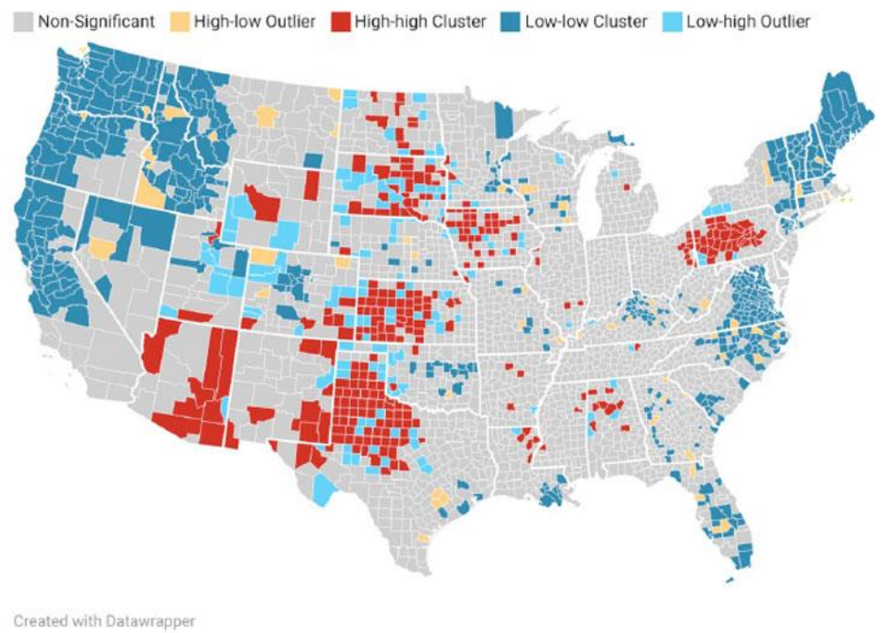


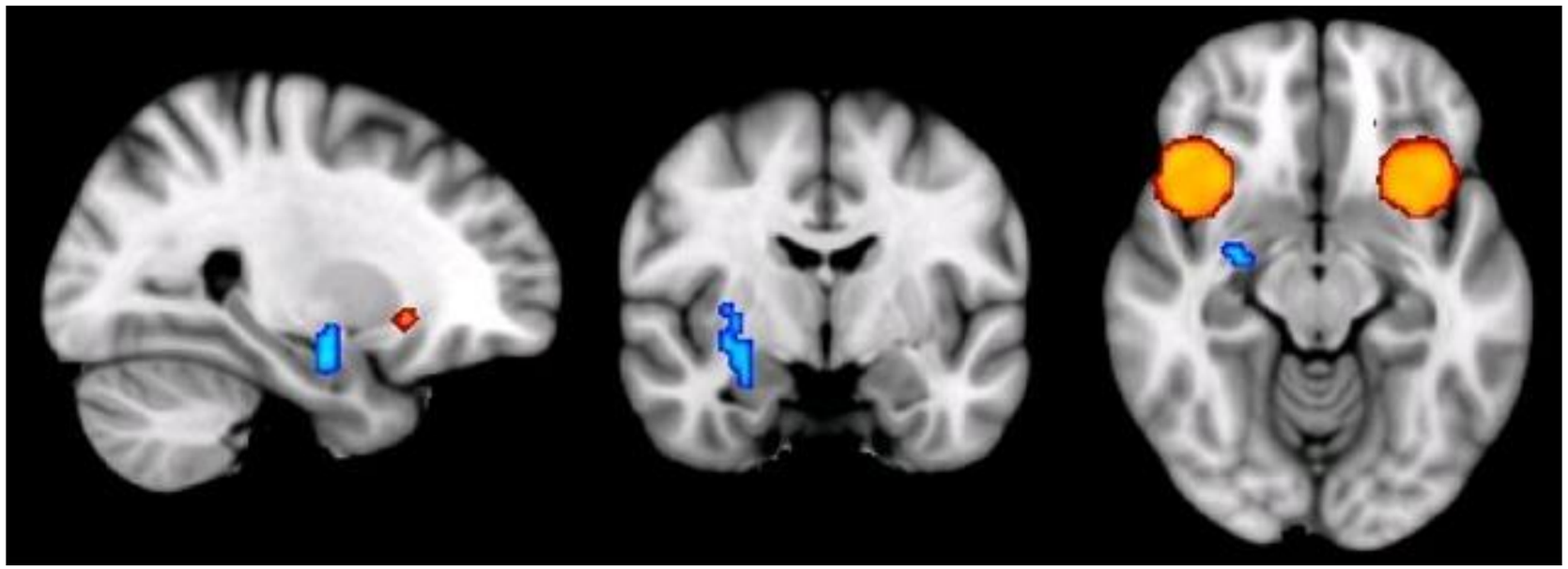
Netflix uses clustering to group similar users and content

- quickly narrow down millions of users/items
- improve recommendation speed
- Provide meaningful group-based suggestions

Other applications



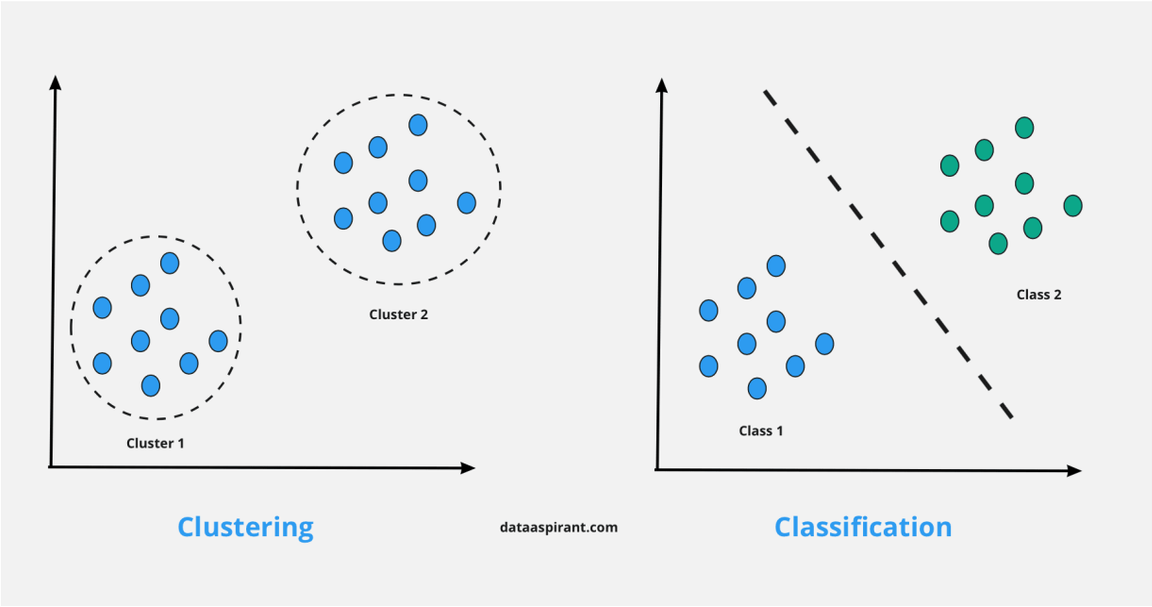




Chavez, Robert S., and Dylan D. Wagner. "Mass univariate testing biases the detection of interaction effects in whole-brain analysis of variance." *BioRxiv* (2017): 130773.

Clustering

Clustering: the task of grouping a set of objects in such a way that objects in the same group (called a cluster) are more similar to each other than to those in other groups



https://dataaspirant.com/4-difference-between-clustering-and-classification/

	Clustering	*Classification*
Number of Classes	Unknown	Known
Training Data	No required	Required
Aim	Work on existing data	Classify future instances into classes

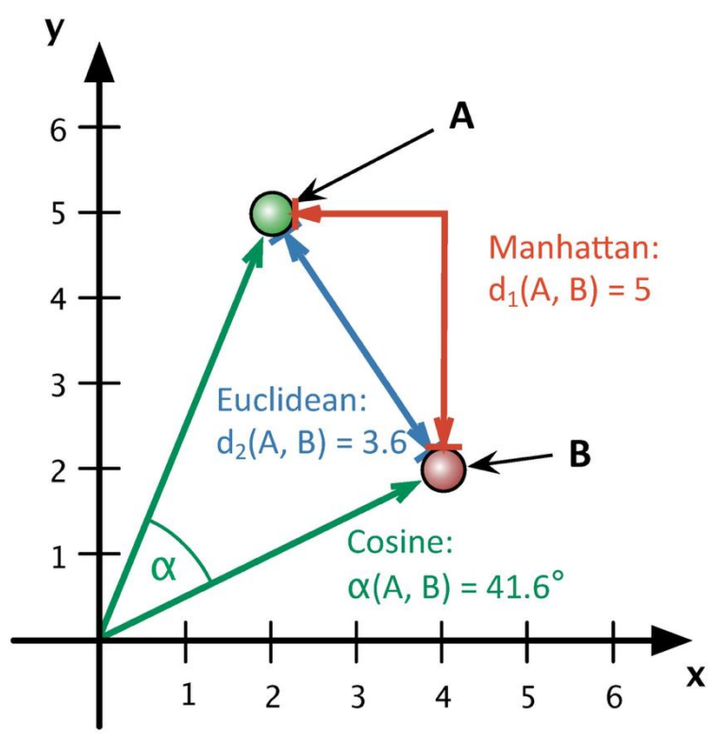
What is similarity?

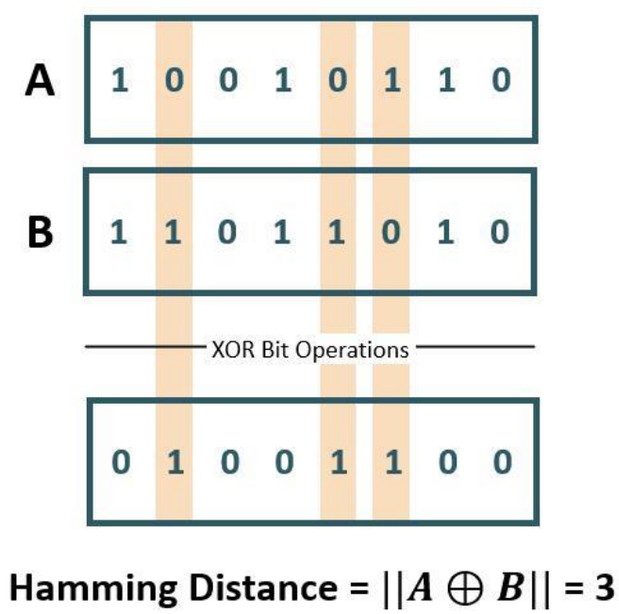
- Similarity is hard to define
- Typically measured by a distance or similarity measure
- Different measures lead to different clusters -> clustering is subjective

Distance measures

- Let O_1 and O_2 be two datapoints from a give dataset. The distance (dissimilarity) between O_1 and O_2 is a real number denoted by $D(O_1,O_2)$

- Depends on the data types
  - Numerical features: Euclidean distance, Manhattan distance, Cosine distance



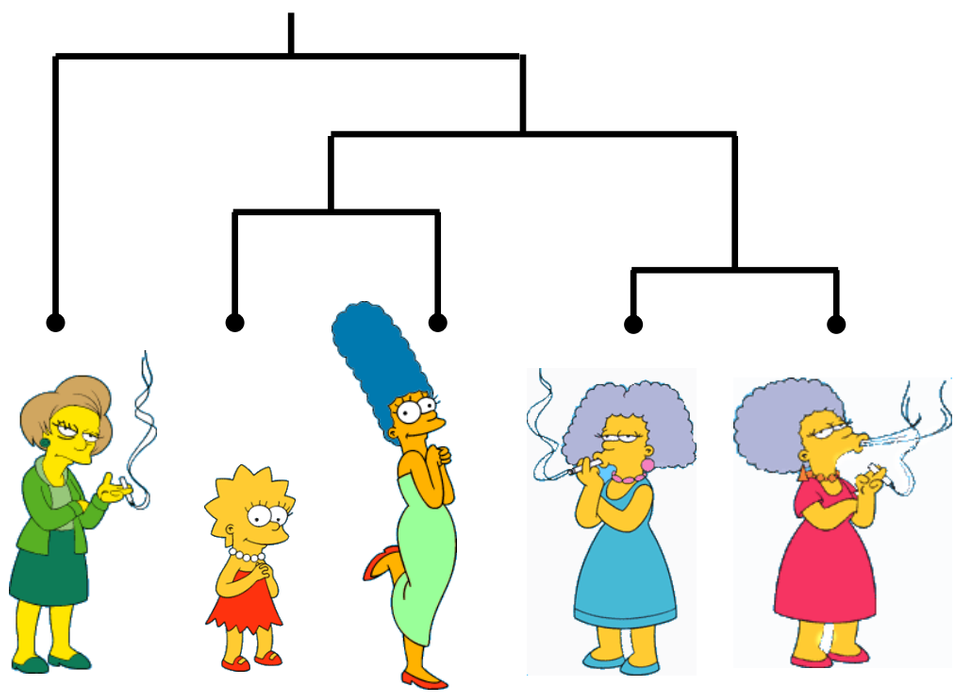


  - Categorical features: Hamming distance

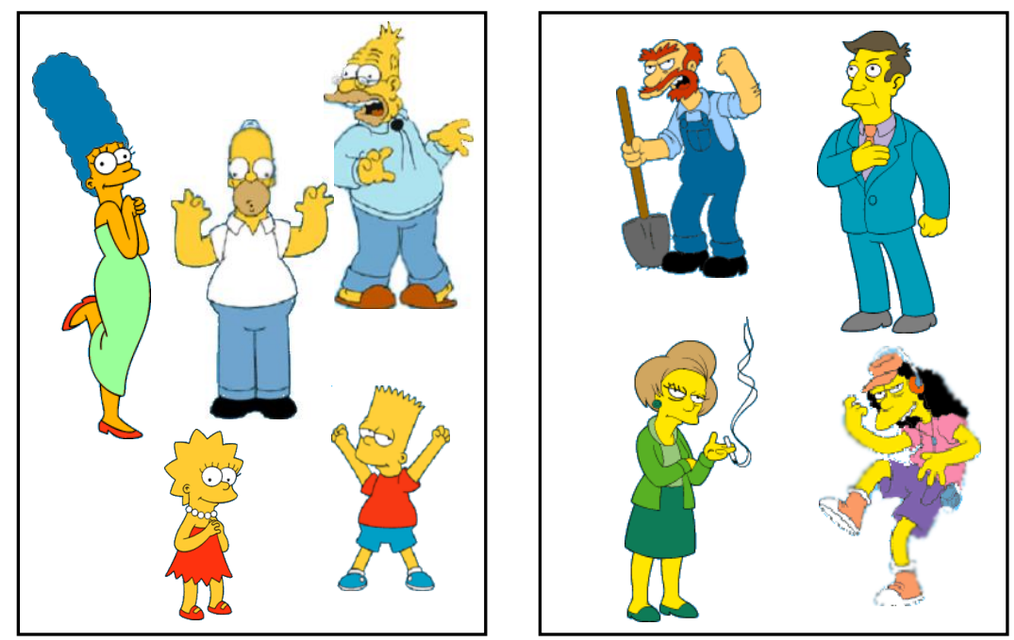
Clustering methods

- Two main types of clustering
- **Partitional algorithms:** Construct various partitions and then evaluate them by a criterion
- **Hierarchical algorithms:** Create a hierarchical decomposition of the set of objects using a criterion

Hierarchical



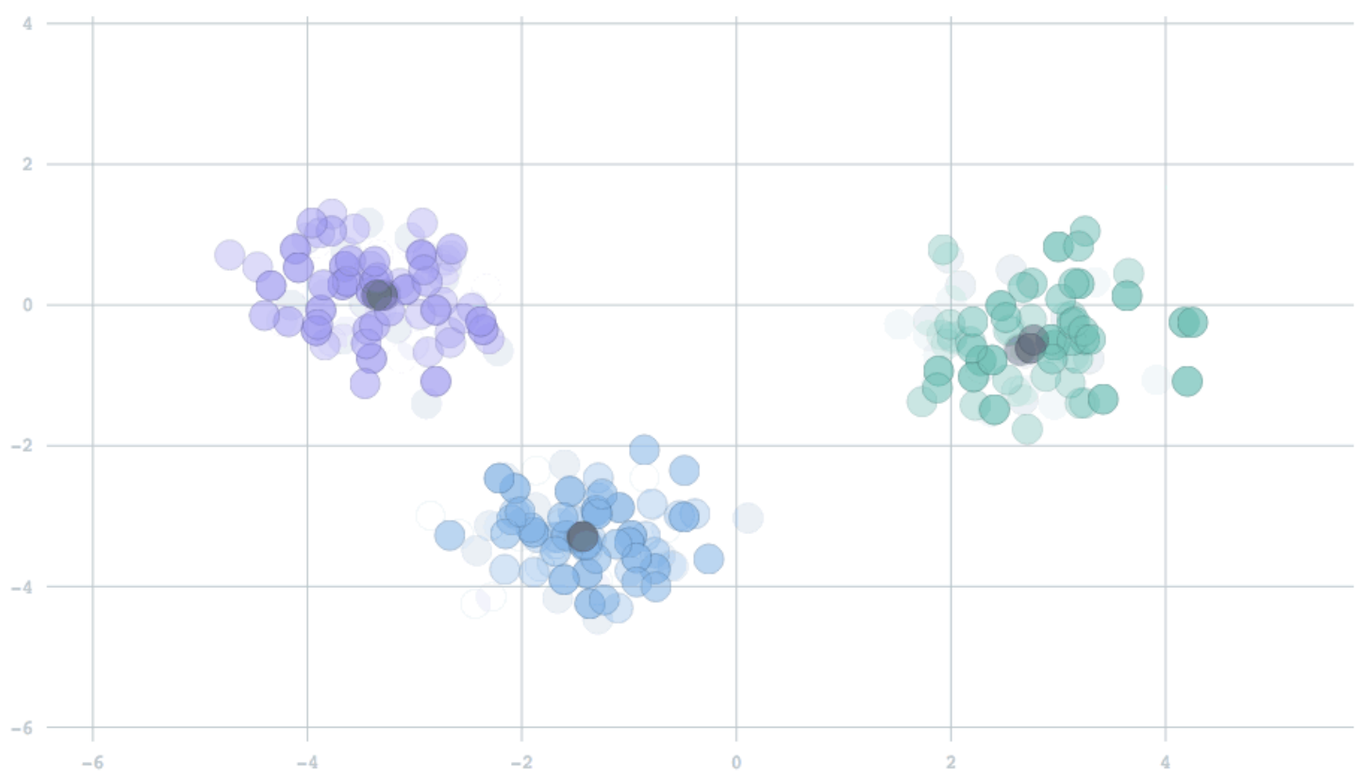
Partitional



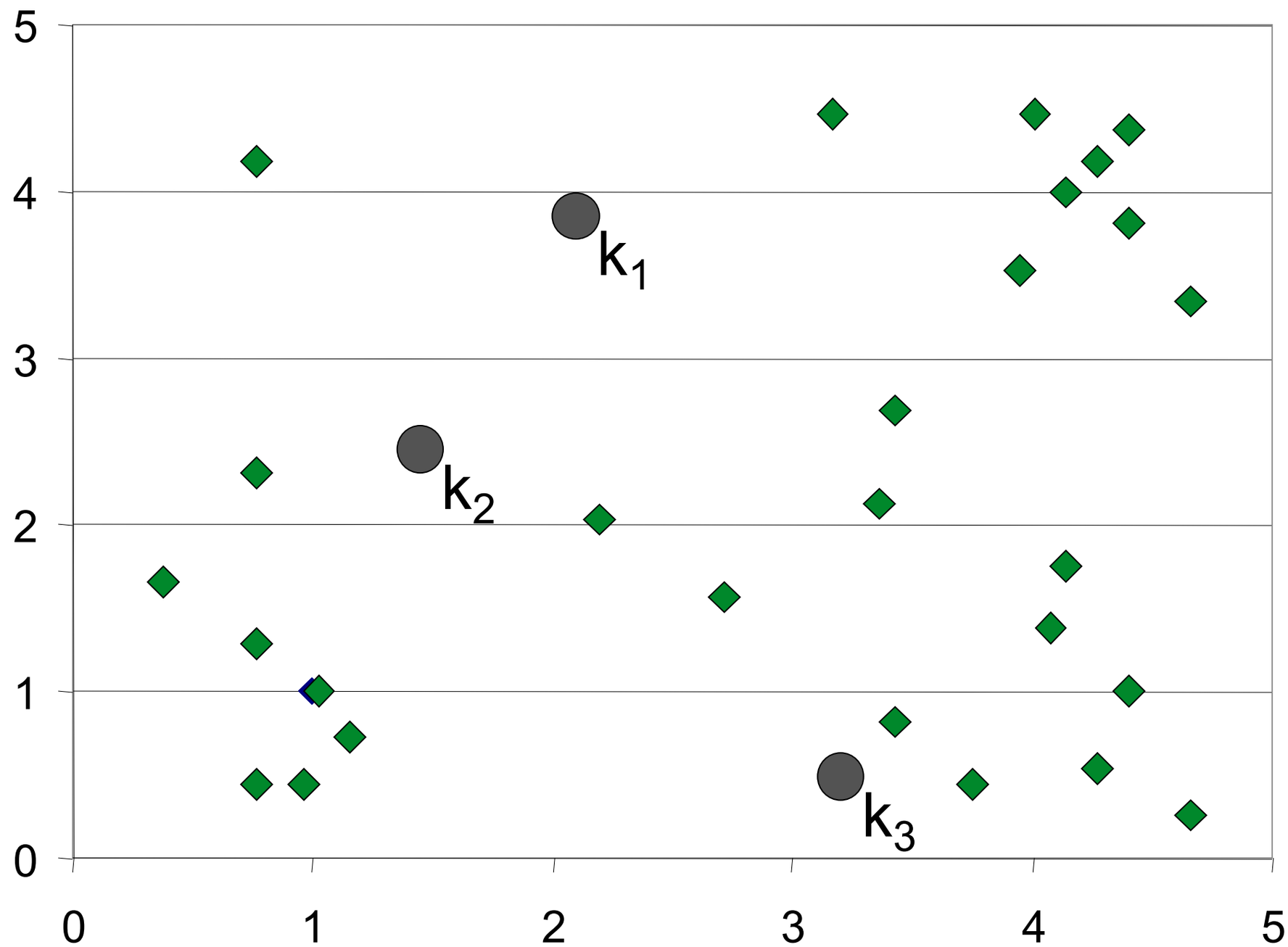
K-Means

- A partitional method

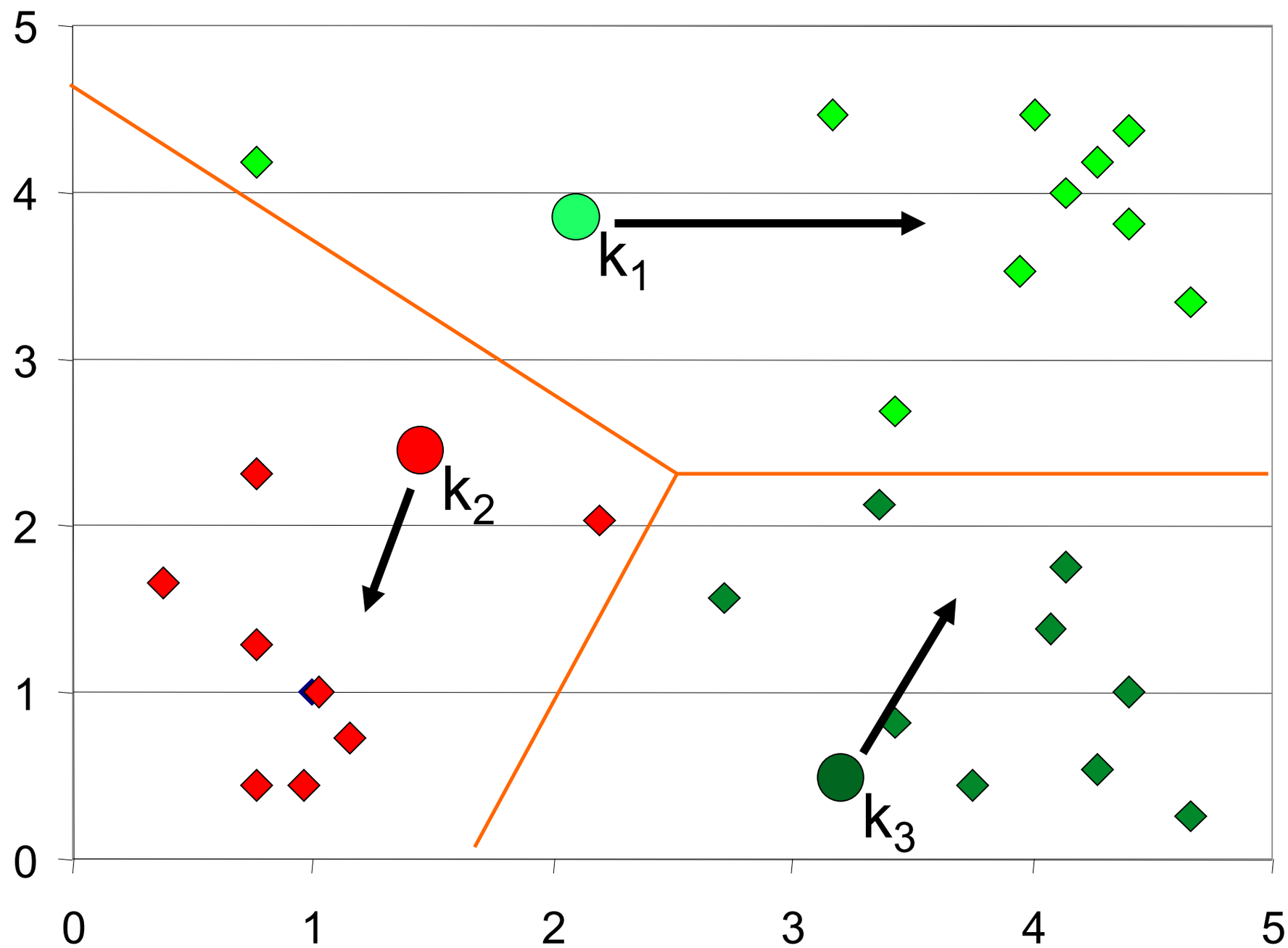
1. Start with *K* random cluster centres, aka centroids
2. Reassign: assign each instance/object to the nearest centroids
3. Updating: compute the new centroid for each cluster as the mean of the objects assigned to the cluster
4. Repeat step 2 until no change to the centroids



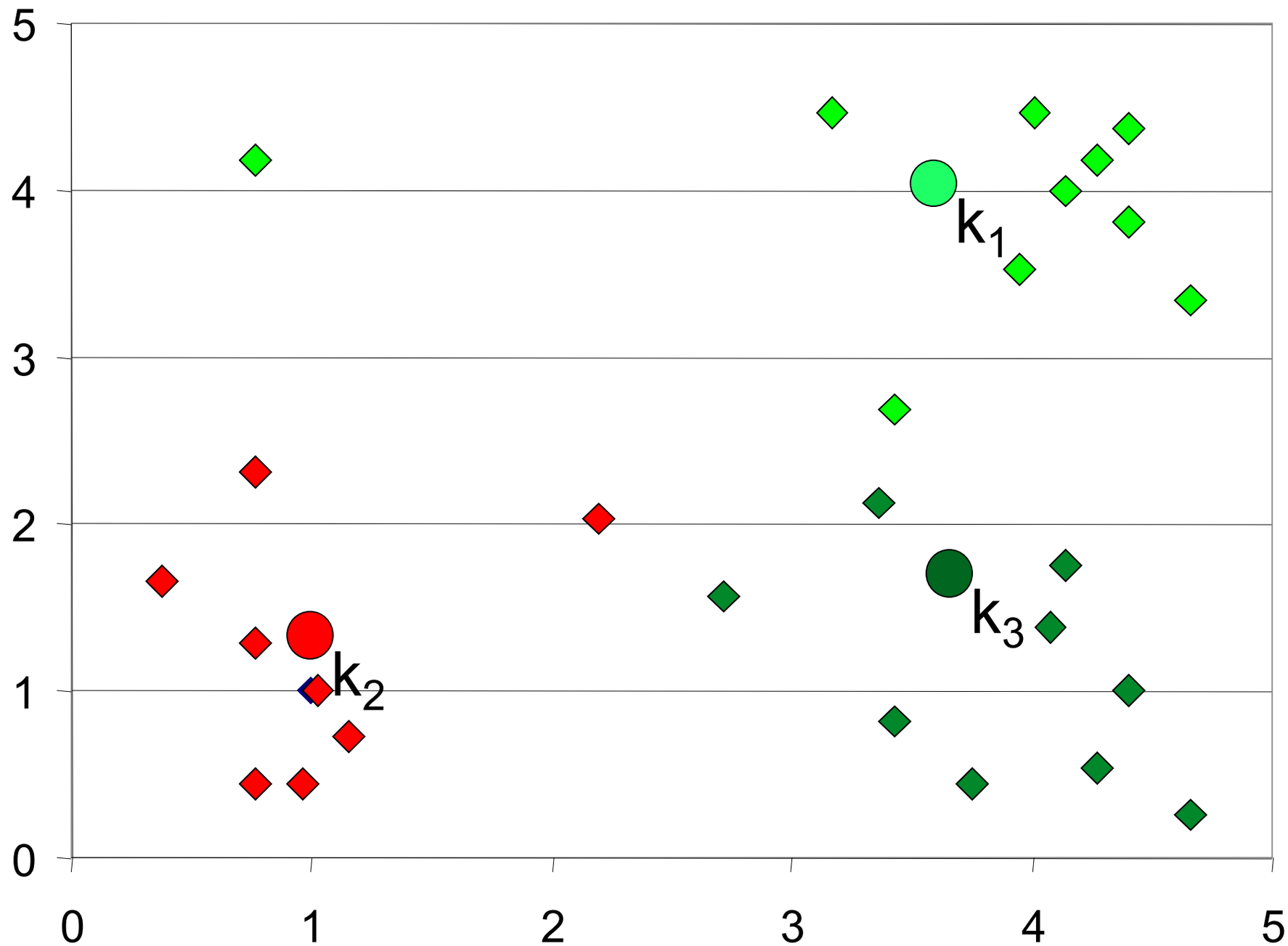
K-means Clustering: Step 1



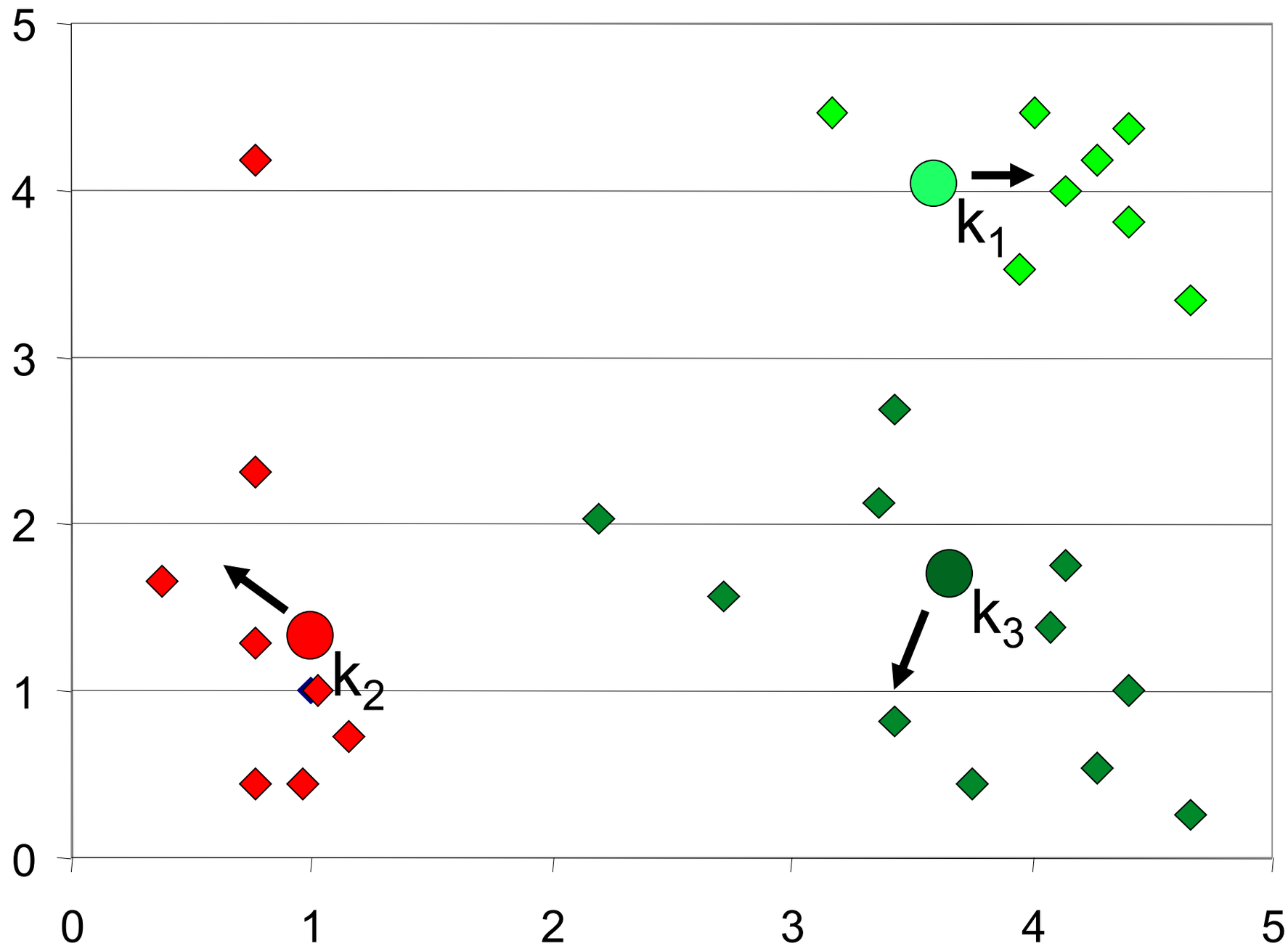
K-means Clustering: Step 2



K-means Clustering: Step 3



K-means Clustering: Step 4



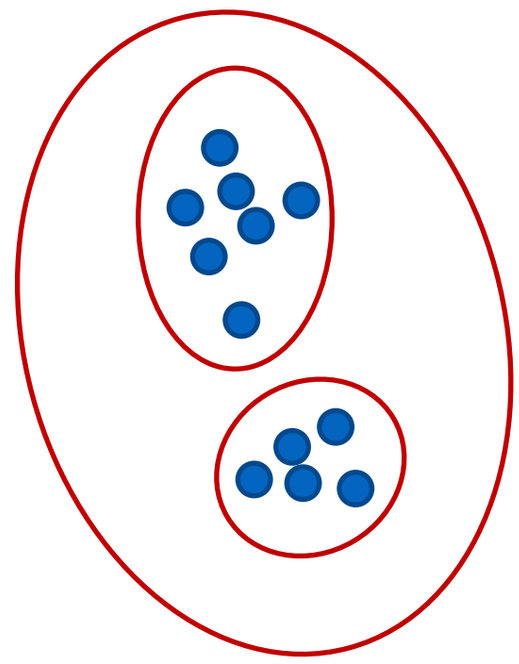
Comments on K-Means

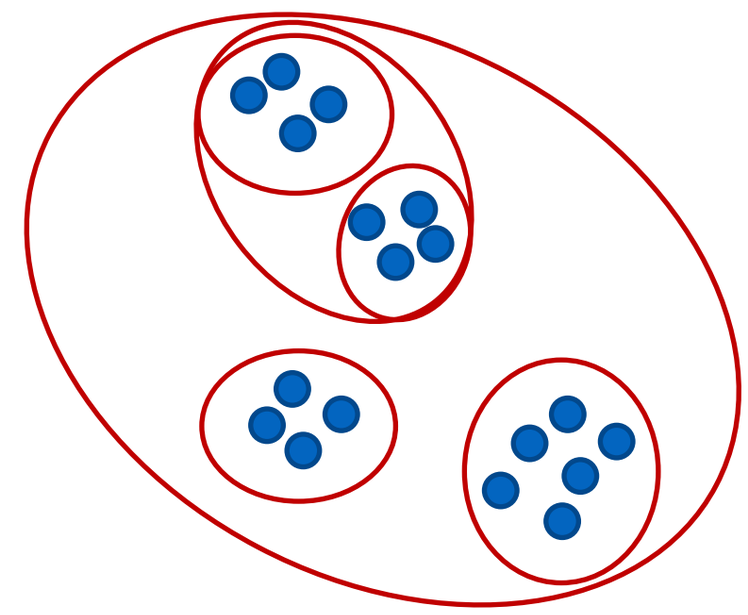
- Very simple and flexible algorithm
- Scale well with large numbers of samples and features

- Limitations:
  - Need to specify *K* in advance
  - Need to re-run to obtain clustering with different numbers of clusters
  - Applicable when mean is defined, what about categorical data?
  - Stochastic algorithm: different initialised centroids -> different clusters
  - Usually convert to local optima

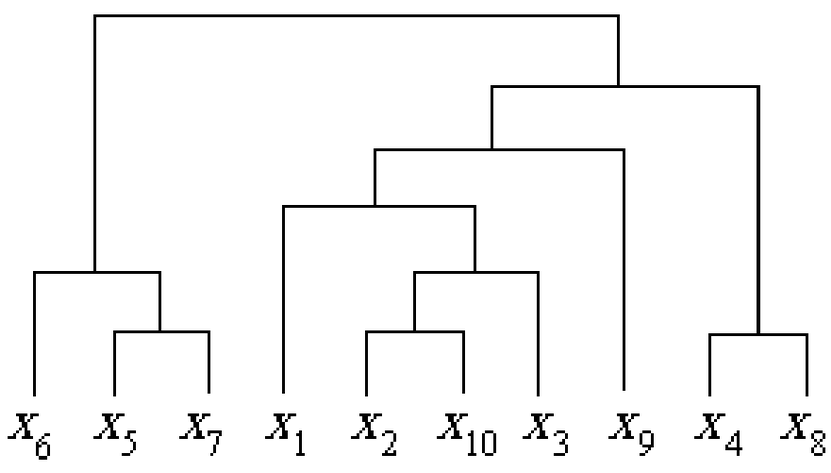
Hierarchical Clustering (1)

- How would you describe data like this?





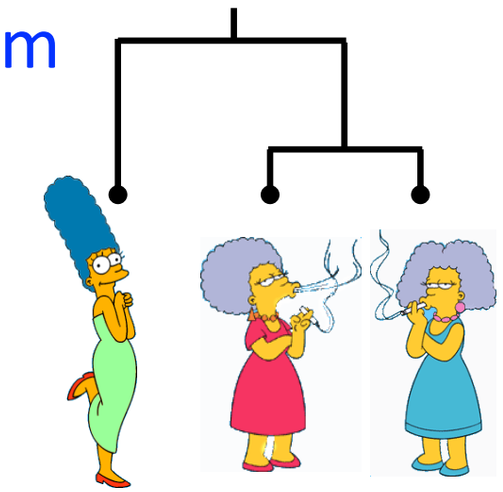
maybe it's "really" a tree?

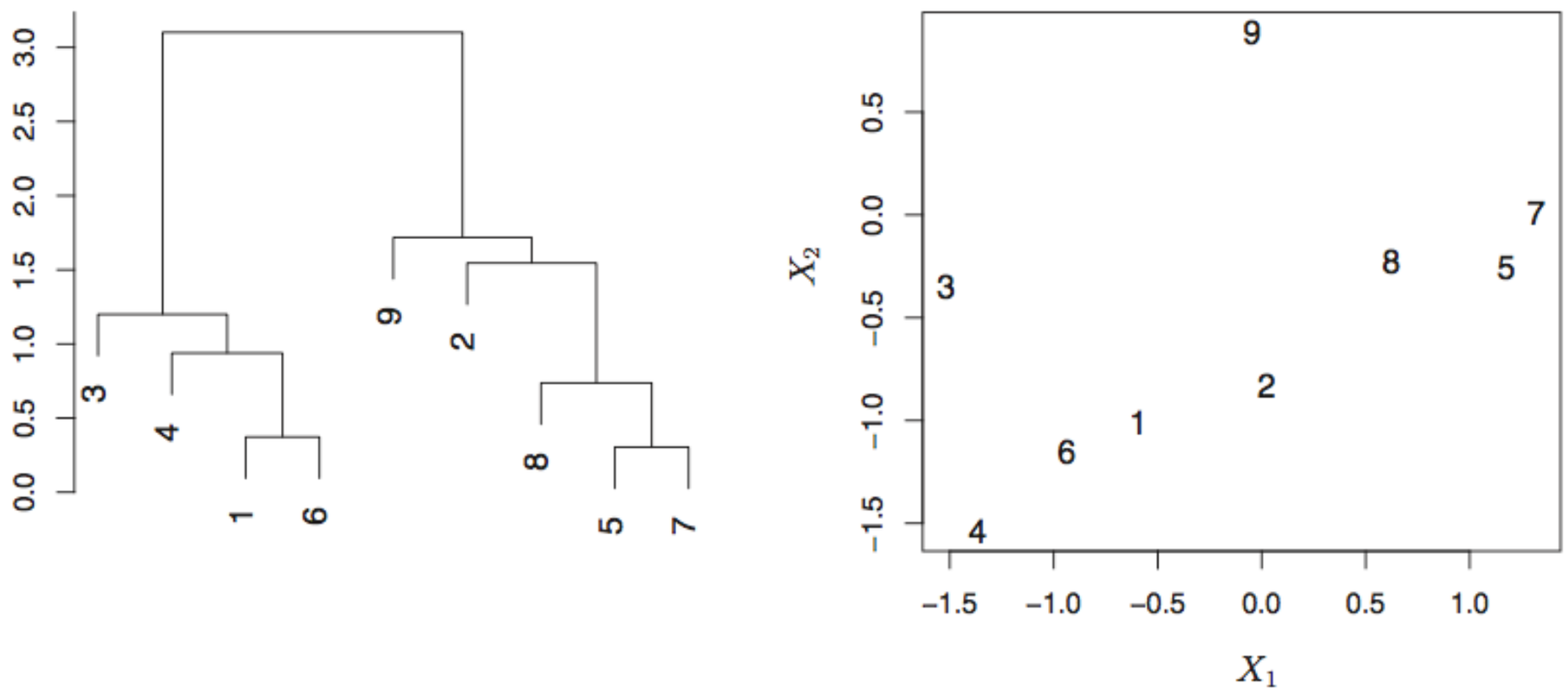


k=2, 5, 6, …

Dendrogram

- The hierarchy of clusters is represented as a tree/dendrogram
- The dissimilarity between two observations is related to the vertical height at which they first get merged into the same cluster. The greater the height, the greater the dissimilarity

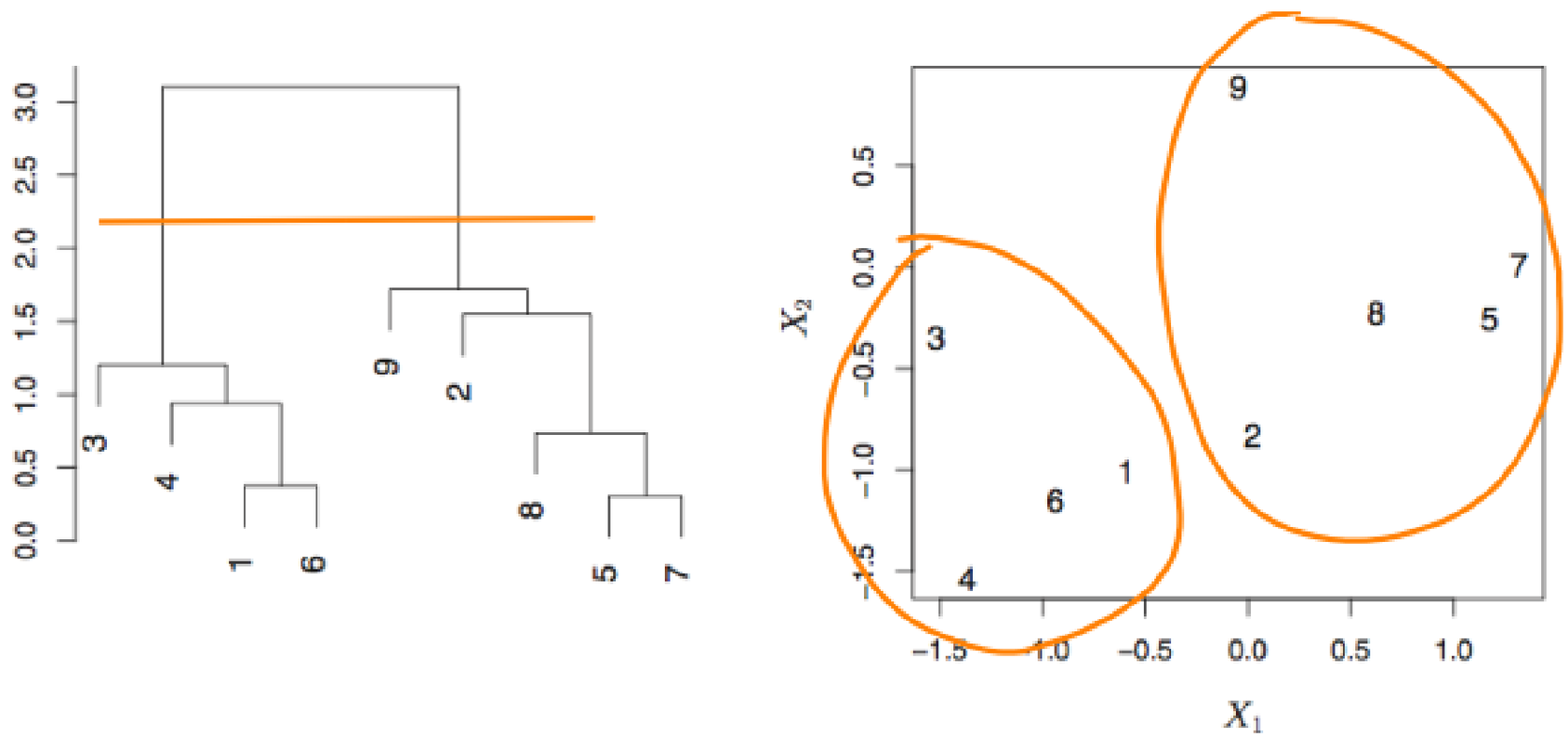




ISLR Figure 10.10: $n = 9$ and $p = 2$

Cutting a dendrogram

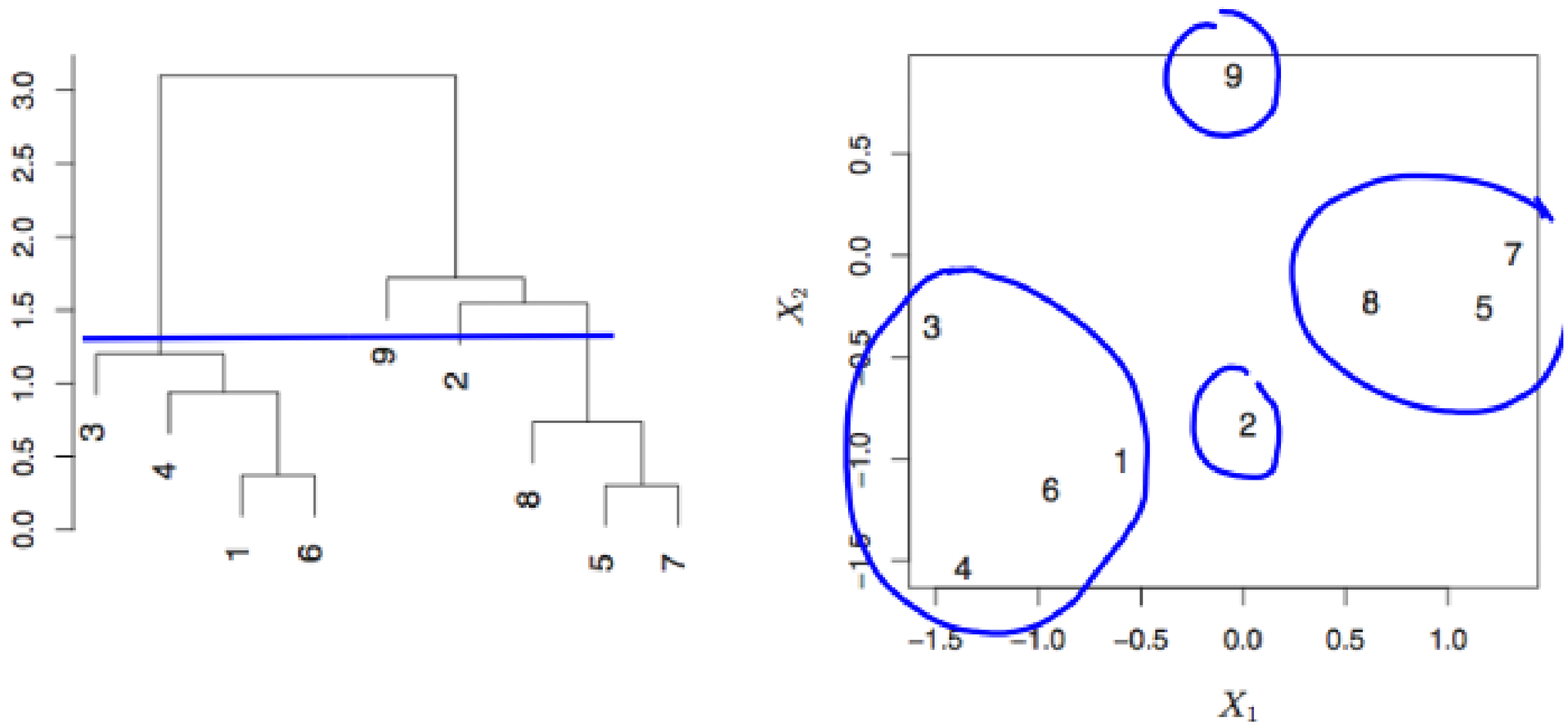
- Cutting a dendrogram horizontally gives a natural clustering. The height of the cut determines the number of clusters
- No need to re-run to get different number of clusters



2 clusters

Cutting a dendrogram

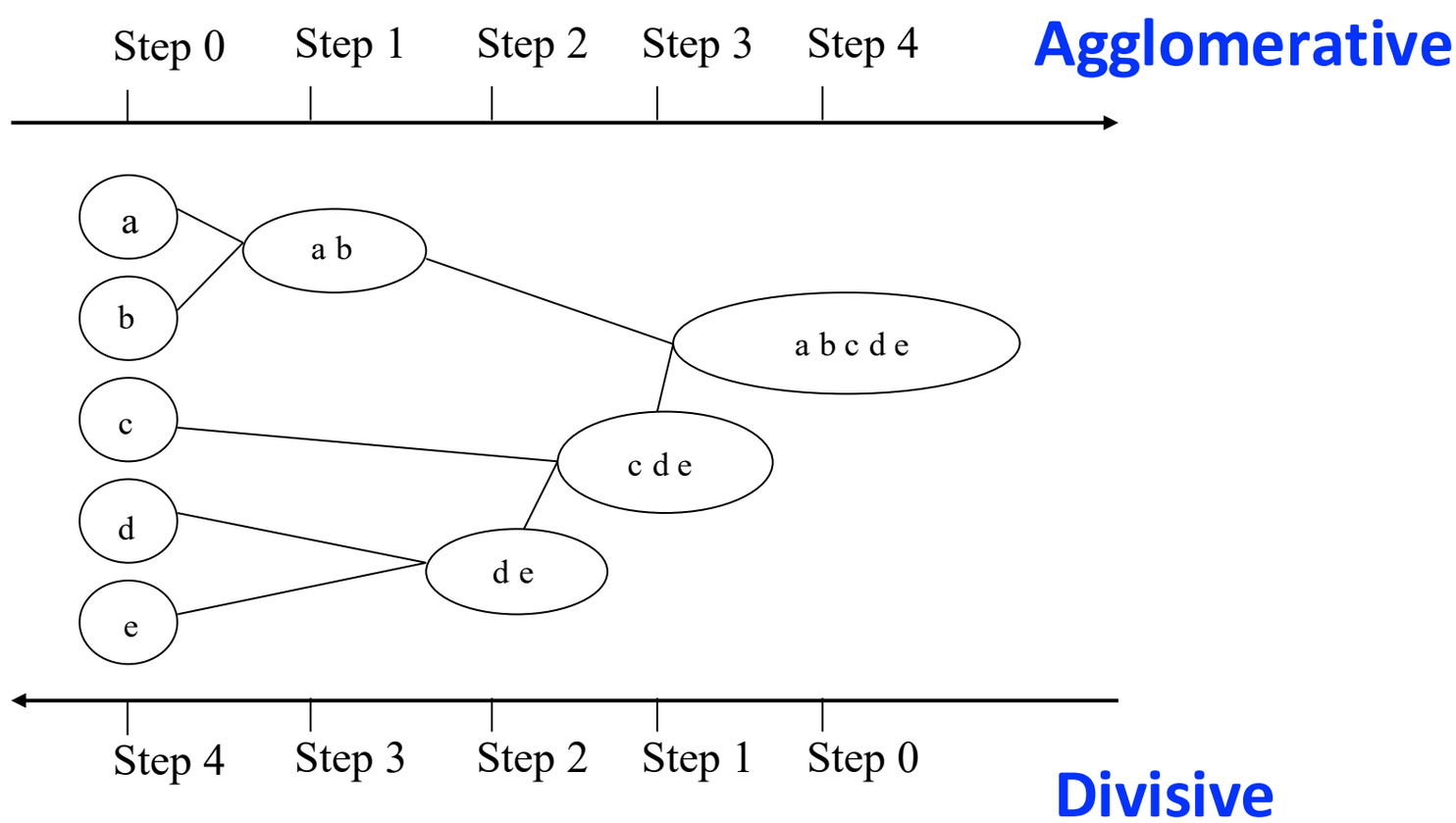
- 4 clusters



4 clusters

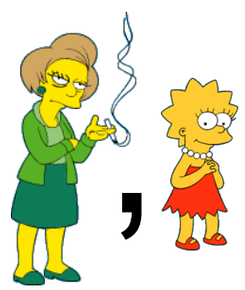
Hierarchical Clustering (2)

- There are two ways to do hierarchical clustering:
  - **Agglomerative** or **bottom-up** clustering where we start with the observations in *n* clusters – the leaves of the tree – and then merge clusters – forming branches – until there is only 1 cluster, the trunk of the tree
  - **Divisive** or **top-down** clustering where we start with the observations in 1 cluster and then split clusters until we reach the leaves
- We will focus on agglomerative clustering as it is generally much more efficient than divisive clustering

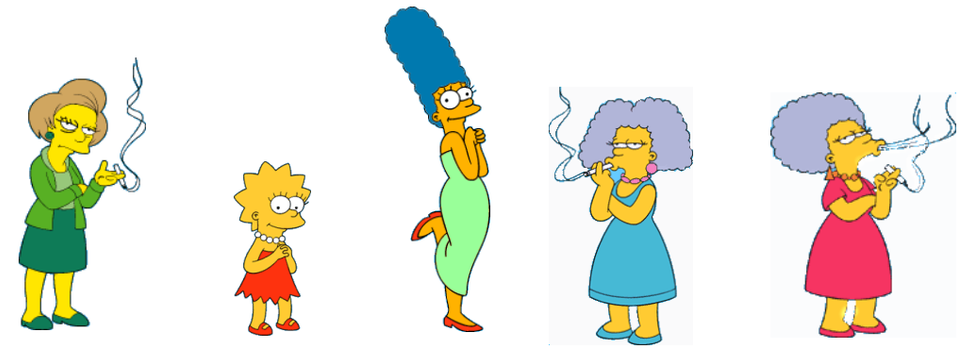


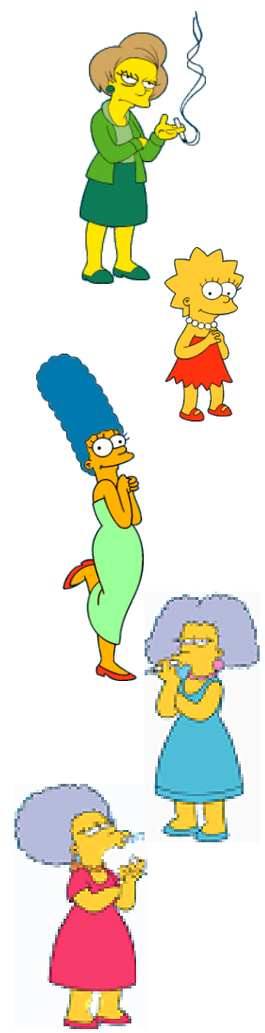
Agglomerative Clustering (1)

We begin with a distance matrix which contains the distances between every pair of objects in our database.

D() = 8

D( ) = 1



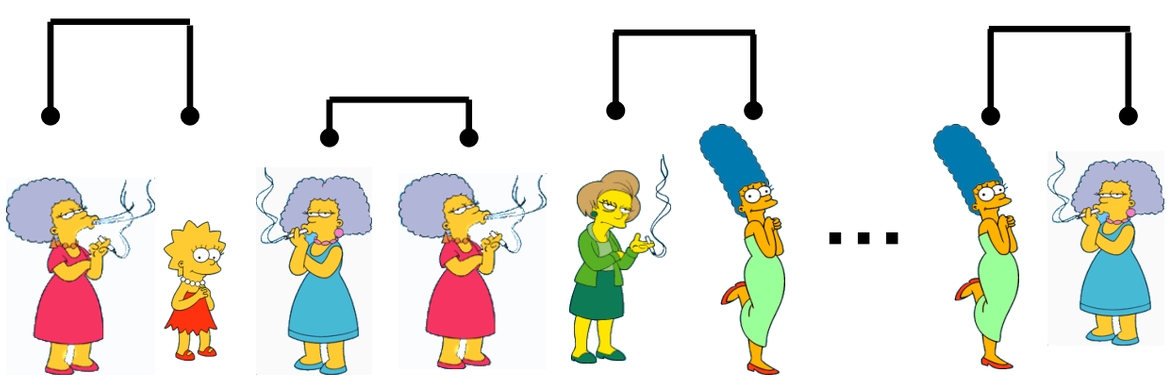


0	8	8	7	7
	0	2	4	4
		0	3	3
			0	1
				0

Agglomerative Clustering (2)

Starting with each item in its own cluster, find the best pair to merge into a new cluster. Repeat until all clusters are fused together.

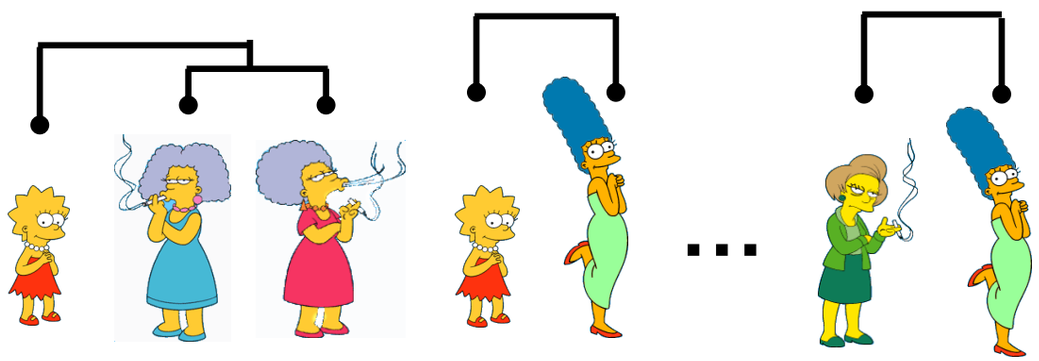
Consider all possible merges…



Choose the best

Agglomerative Clustering (3)

Consider all possible merges…



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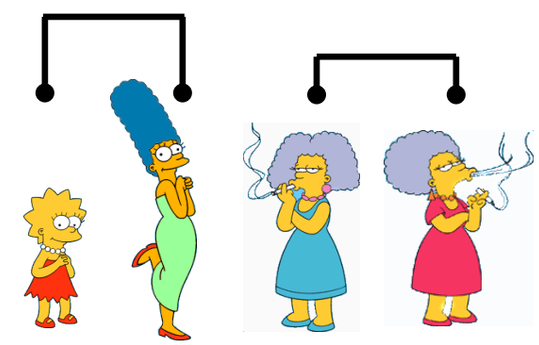

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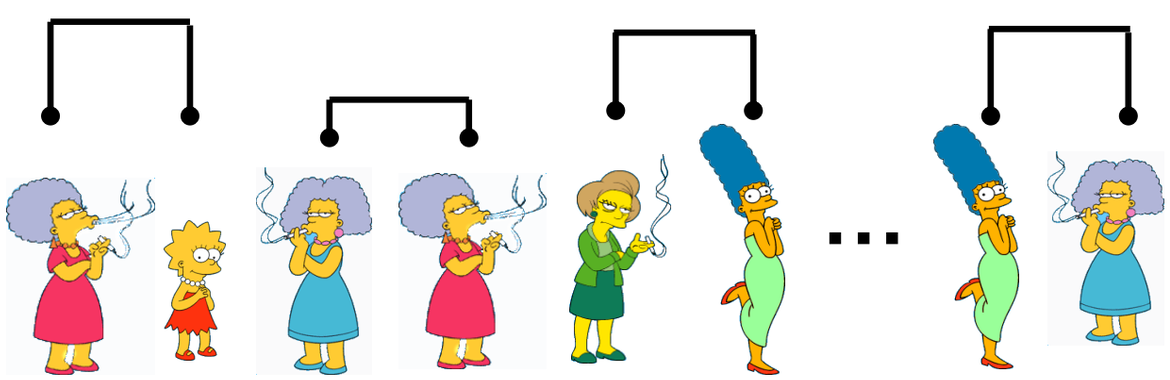

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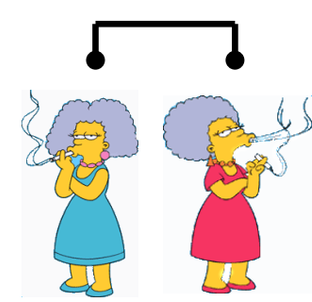
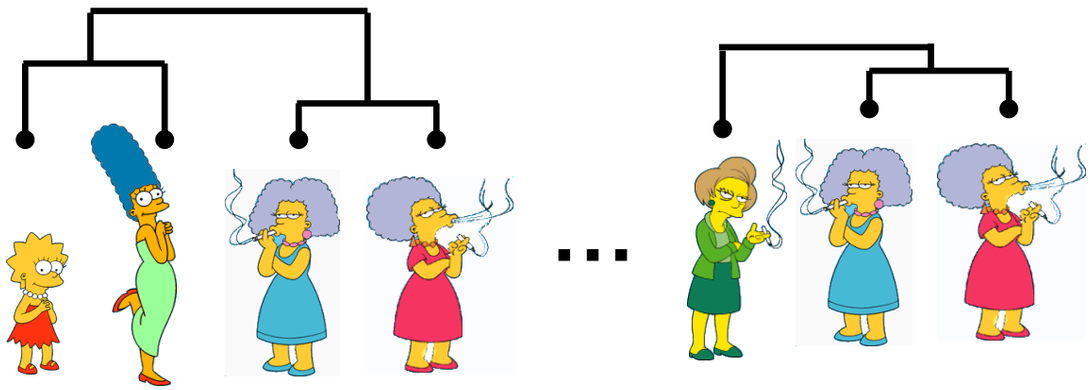


Agglomerative Clustering (4)

Consider all possible merges…



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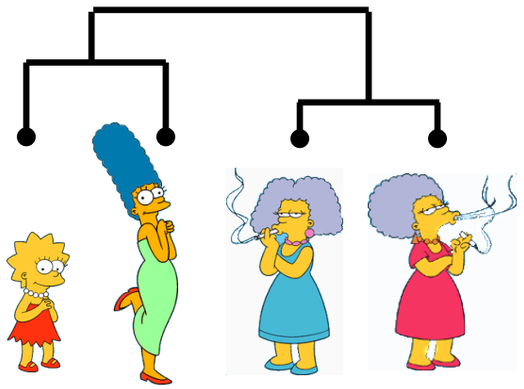

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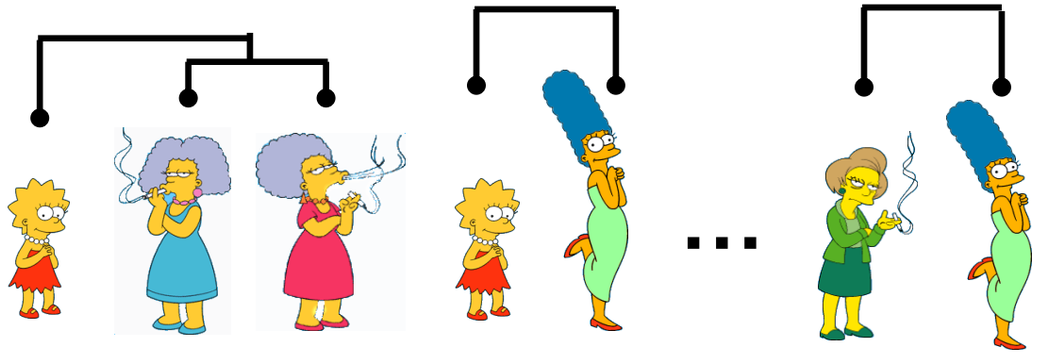

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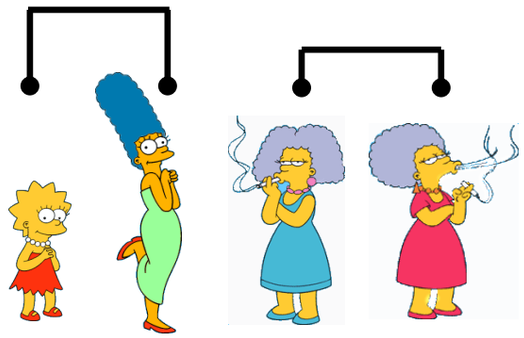

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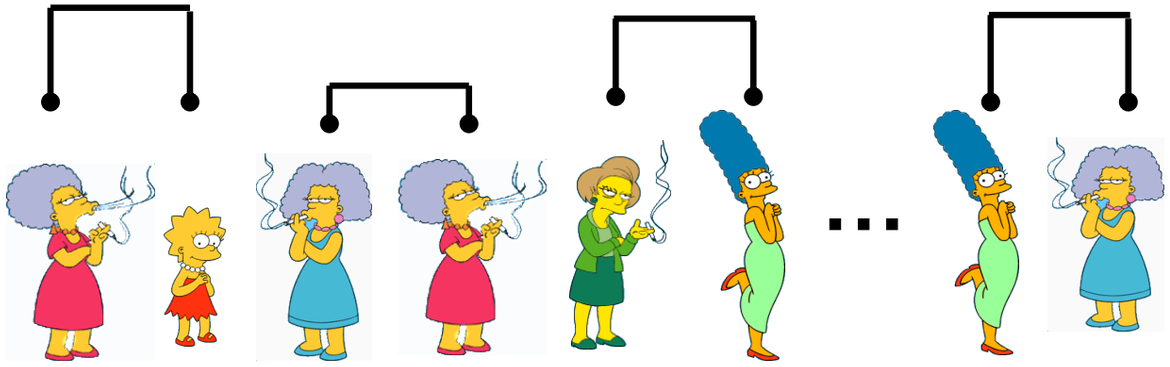
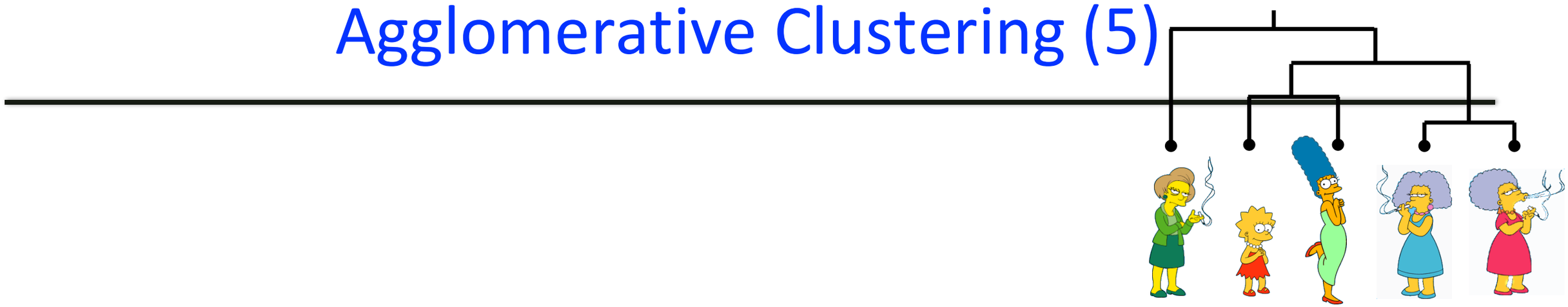


Choose the best

Agglomerative Clustering (5)



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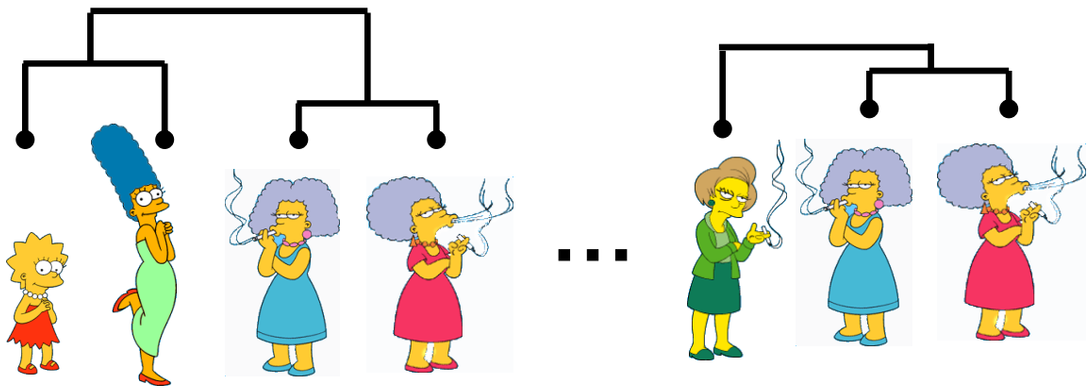

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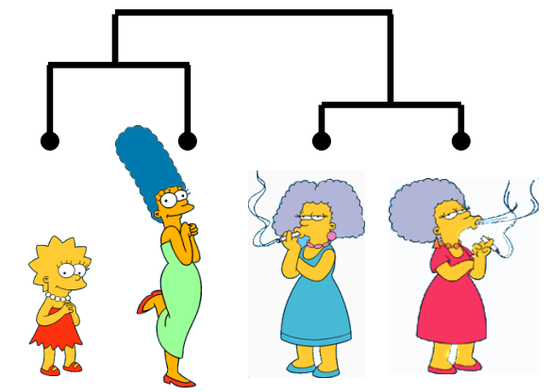

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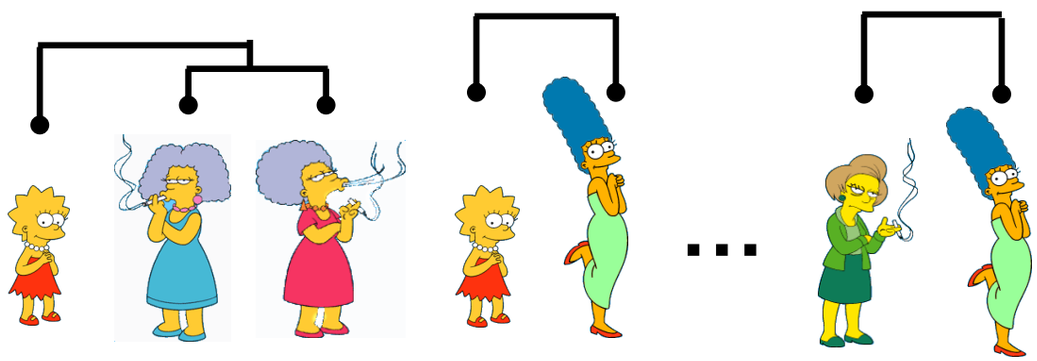

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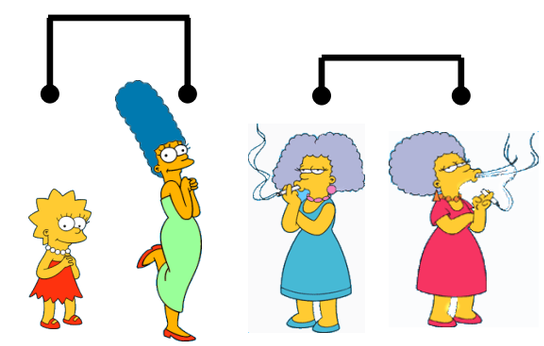

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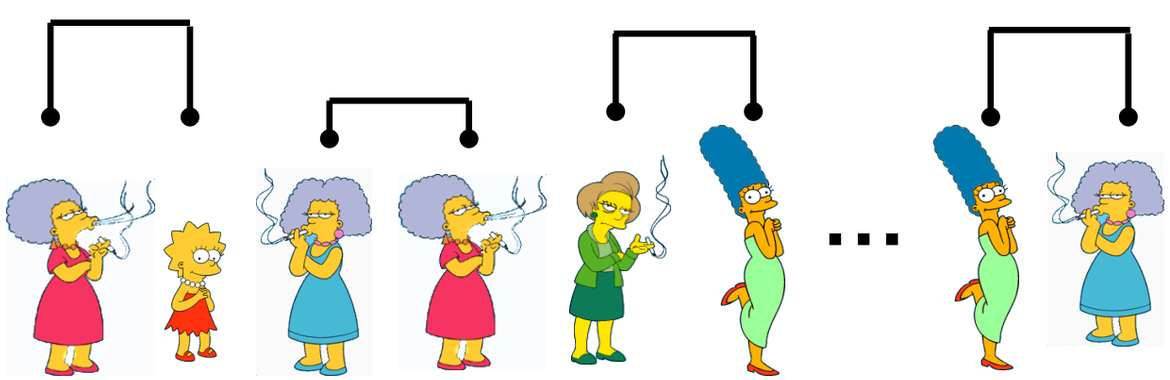

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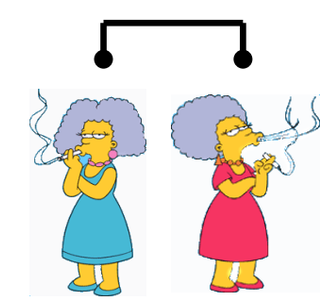
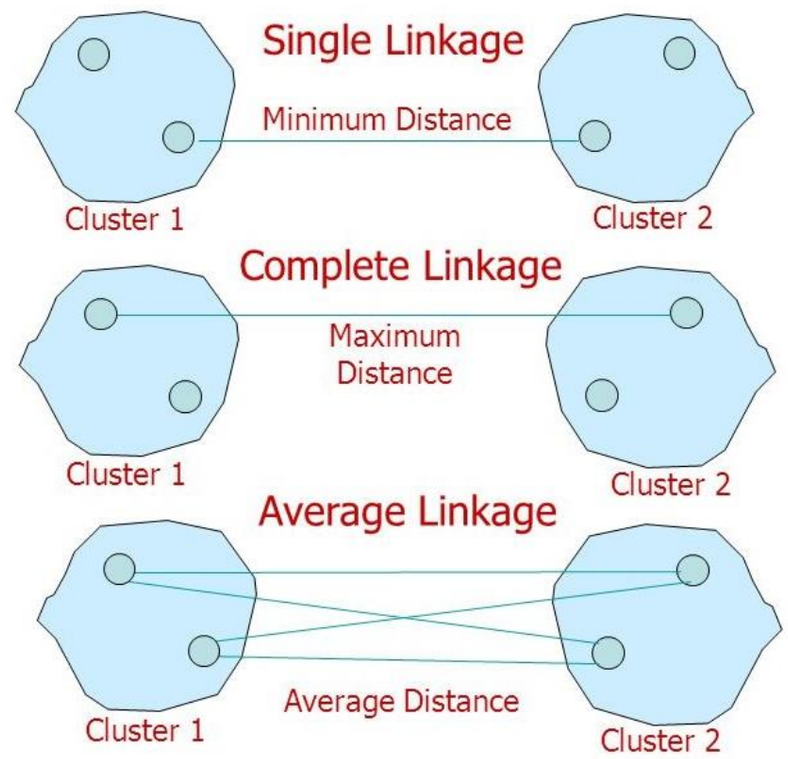


Linkage Methods (1)

- Agglomerative Clustering merges clusters based on distance between clusters – defined by linkage method

- **Single**: compute the minimum pairwise dissimilarity where one observation is in cluster A and the other is in cluster B.
- **Complete**: compute the maximum pairwise dissimilarity where one observation is in cluster A and the other is in cluster B
- **Average**: compute the average pairwise dissimilarity where one observation is in cluster A and the other is in cluster B



Linkage Methods (2)

Generally,

- Complete and average linkage produce more balanced dendrograms;
- Single linkage can produce trailing clusters in which single observations are merged one-at-a-time.

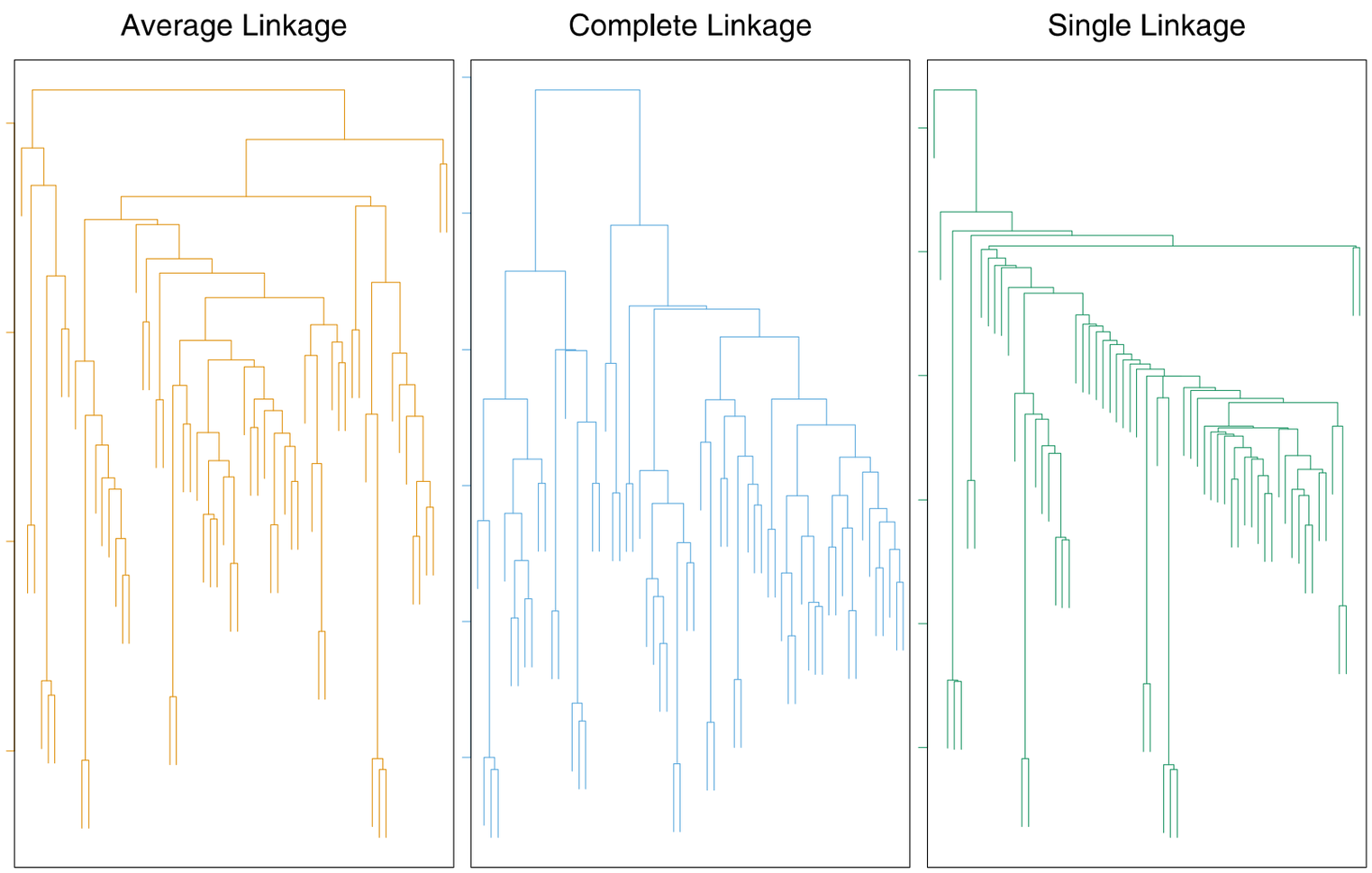


FIGURE 10.12. *Average, complete, and single linkage applied to an example data set. Average and complete linkage tend to yield more balanced clusters.*

Agglomerative clustering algorithm

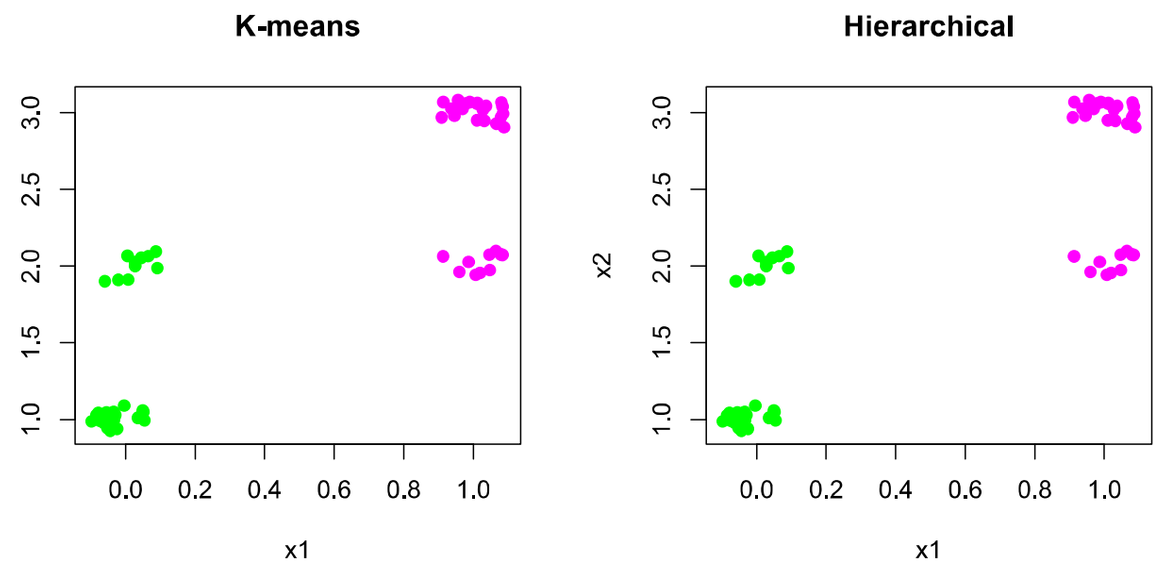
With the choice of dissimilarity measure and linkage method, **agglomerative clustering proceeds** as follows:

- Treat each observation as its own cluster, *n* clusters. Compute all pairwise dissimilarities (such as Euclidean distance) of all the $\binom{n}{2} = \frac{n(n-1)}{2}$ pairwise dissimilarities.
- For $i = n, n-1, \ldots 2$
  - (a) Find the pair of clusters that are the least dissimilar and merge them
    - The dissimilarity between these two clusters indicates the height on the dendrogram where the merge is shown.
  - (b) Compute all pairwise dissimilarities between the *(i-1)* remaining clusters

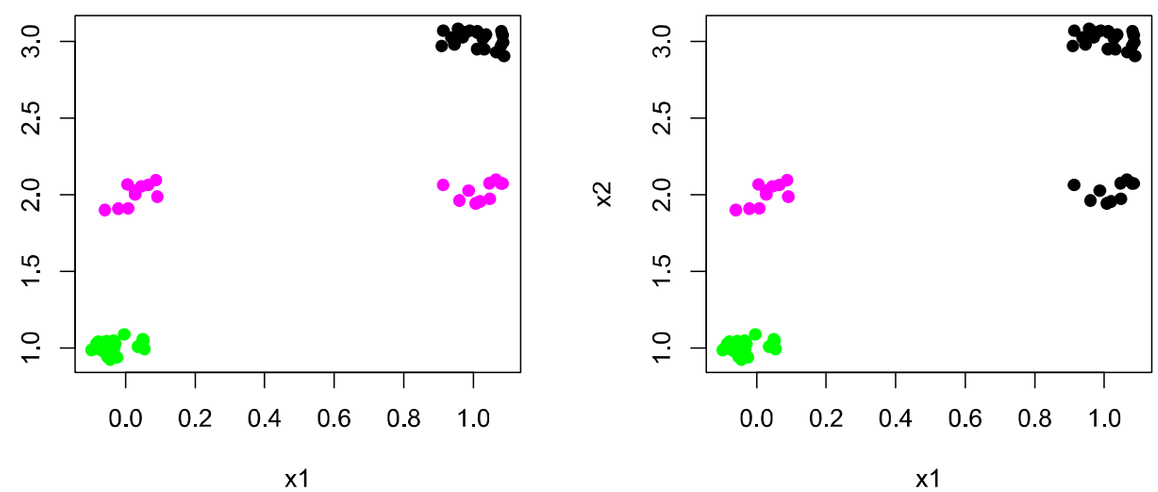
Note that there is no random initialisation, so agglomerative clustering is a deterministic algorithm

Comments on Agglomerative clustering

- Do not need to specify *K* in advance
- Do no need to re-run to obtain clustering with different numbers of clusters
- Applicable to categorical data
- Deterministic algorithm



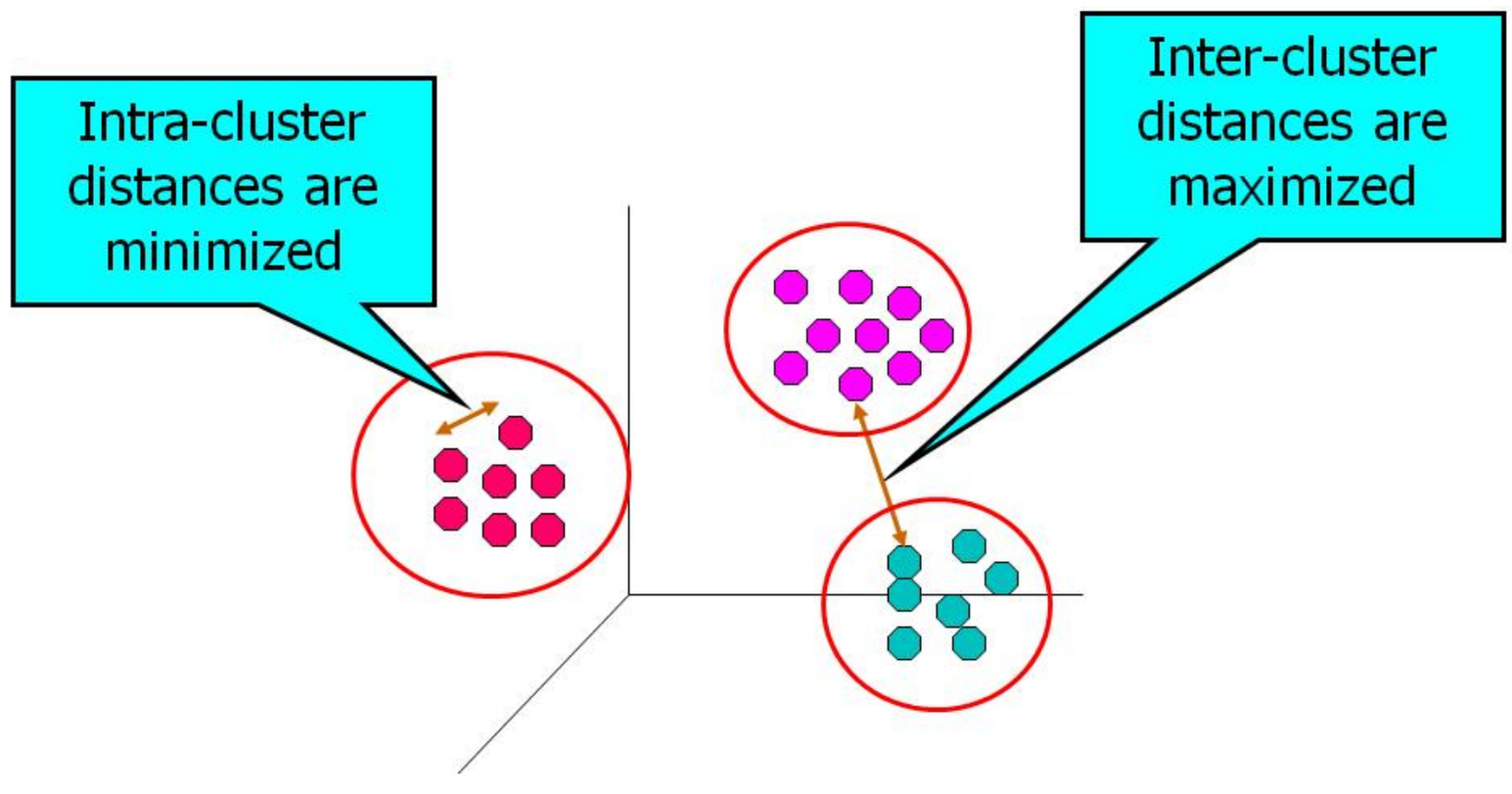
- A *potential drawback* of hierarchical clustering is that clustering obtained by cutting the dendrogram at a certain height is necessarily ***nested*** within the clustering obtained by cutting at a greater height
- Computationally expensive given a large number of samples due to pair-wise distances



Clustering performance

- **Compactness**: how tightly-packed a cluster is.
  - Clusters should be as compact as possible, so as to ensure that only the most related/similar instances have been grouped together.
  - Measured by intra-cluster distance - minimised

- **Separability**: how well neighbouring clusters are separated in the feature space.
  - Measured by inter-cluster distance - maximised



Silhouette Score

$$Silhouette(i) = \frac{b(i) - a(i)}{max\{a(i), b(i)\}}$$

- *a(i)* is the *average* distance between instance *i* and all other instances in its cluster;
- *b(i)* is the *minimum* average distance between instance *i* and the instances in each other cluster.

- Measures how well a given instance is matched to its cluster
- The average silhouette computed across all instances in a partition gives a measure of how good the partition is
- Implicitly balances both the intra- and inter-cluster metrics.

- 1 indicates an instance is perfectly clustered
- –1 indicates it should be in a neighbouring cluster;
- 0 indicates it is on the border of two clusters

Other metrics

- Davies-Bouldin index
- Dunn index
- Calinski-Harabasz Index
- ….

- https://scikit-learn.org/stable/modules/clustering.html#clustering-performance-evaluation

Density Based Clustering

- Density-based clustering algorithms are based on the idea that clusters are dense regions of data, separated by low-density regions
- They are particularly useful for data that contains noise and outliers.
- They can find clusters of arbitrary shapes and sizes.
- We will look at Mean Shift and DBSCAN, but there are many other density-based clustering algorithms (OPTICS, HDBSCAN)

Mean Shift

a sliding-window-based algorithm that attempts to find dense areas of data points

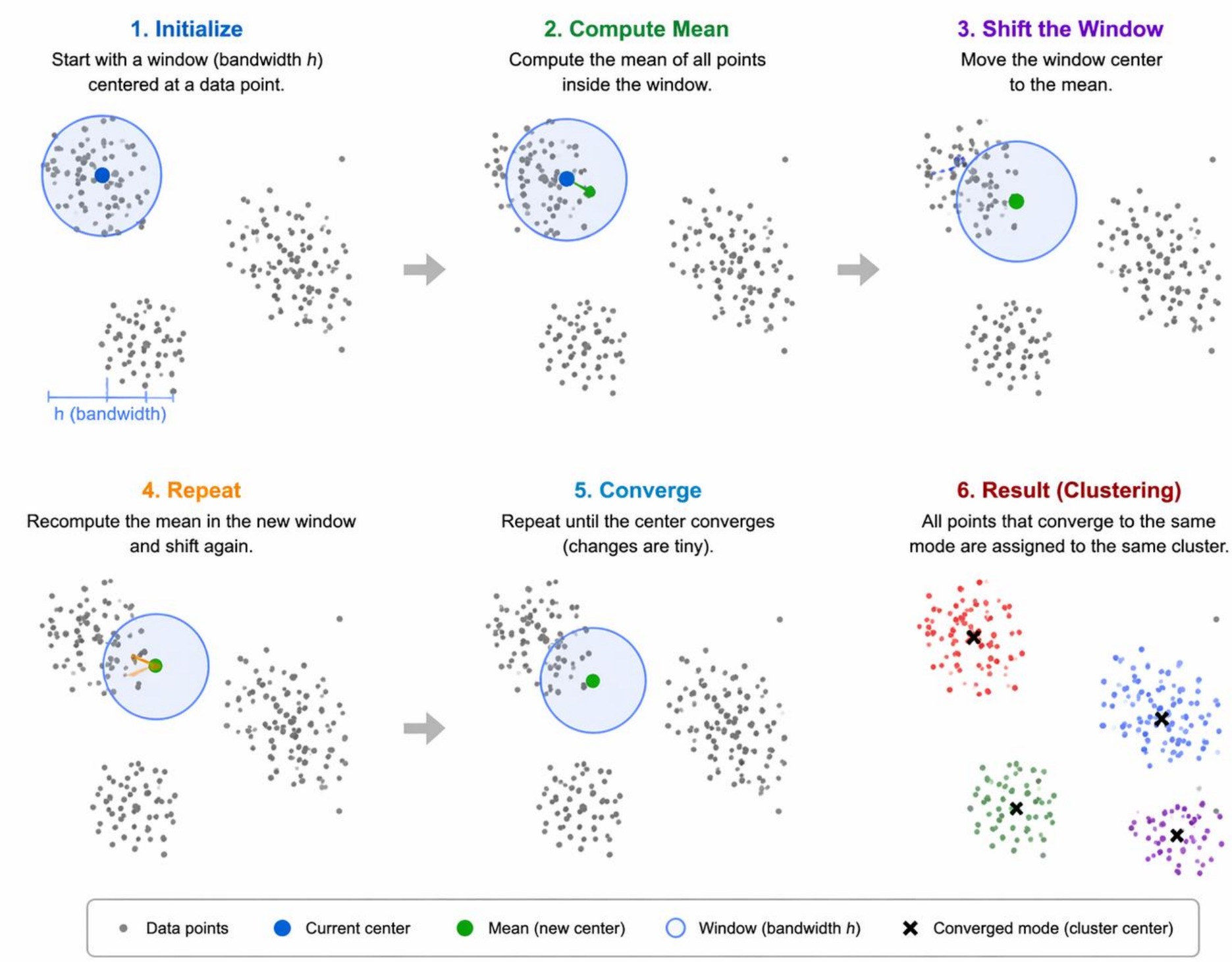
- a density-based method that finds dense regions (modes)
- uses sliding window (bandwidth) over data
- representative points move toward higher-density areas
- no need to specify number of clusters

https://towardsdatascience.com/the-5-clustering-algorithms-data-scientists-need-to-know-a36d136ef68

How Mean Shift Works

Algorithm (Intuition)

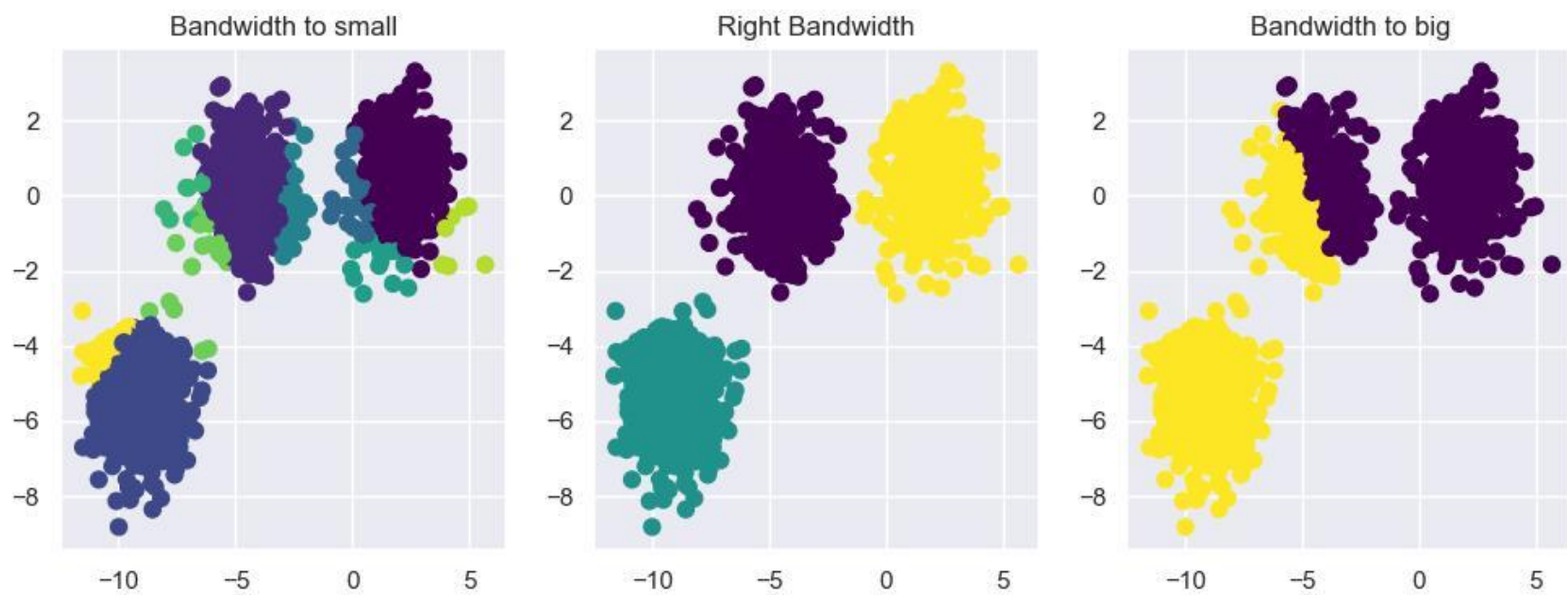
- 1. start from a data point
- 2. compute mean of nearby points (within bandwidth)
- 3. shift the representative points (in blue) toward the mean (in green)
- 4-5. repeat until convergence



Key Points of Mean Shift

- ✓ Finds clusters of arbitrary shapes
- ✓ Automatically determines number of clusters

- ✕ Sensitive to bandwidth choice
- ✕ Computationally expensive



💡 Interpretation:

- Mean Shift = moving points uphill in density until they reach a peak

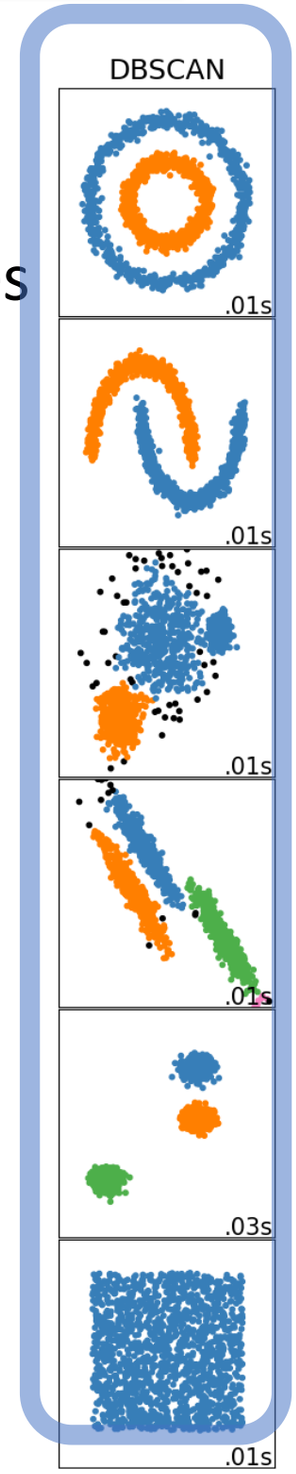
DBSCAN

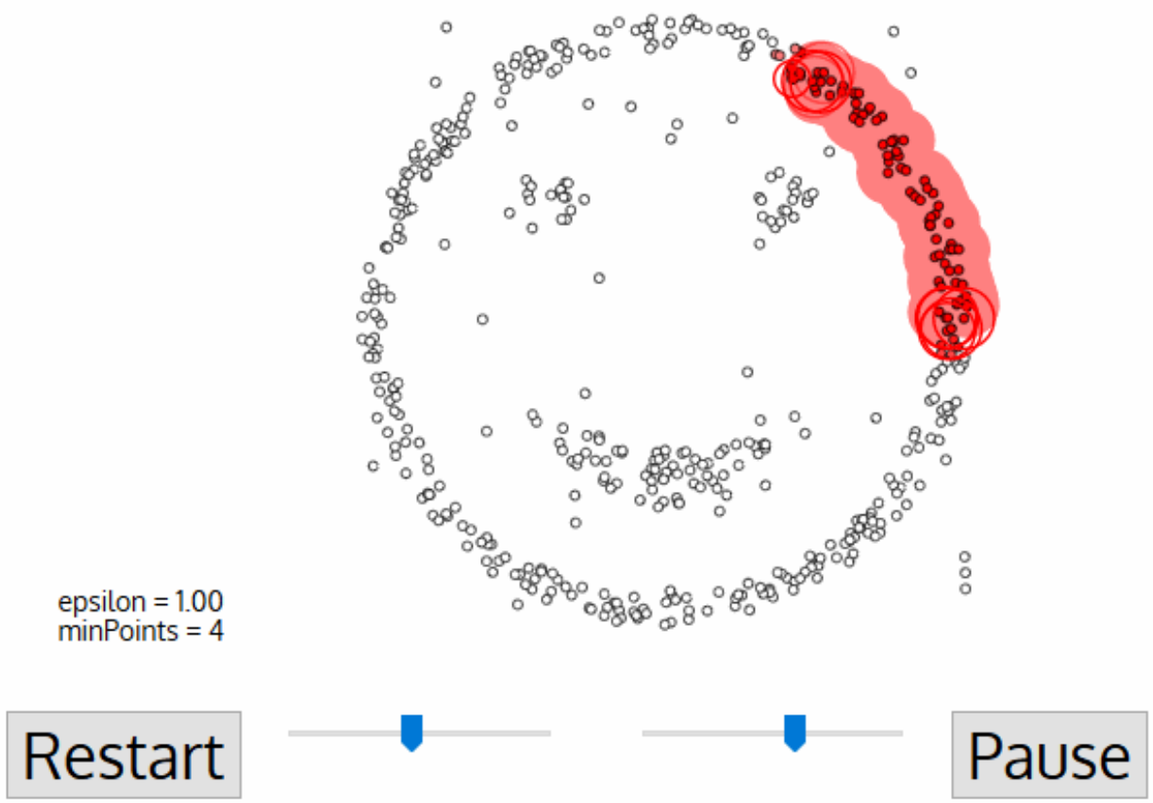
Density-based spatial clustering of applications with noise (DBSCN)

- a density-based clustering techniques, views clusters as areas of high density separated by areas of low density
- basic idea: a point belongs to a cluster if it is close to many points from that cluster
- two key parameters:
  - eps - the distance that specifies the neighborhoods
  - minPts - minimum number of data points to define a cluster
- Points are identified as: core samples/points, which are samples that are in areas of high density; or outliers

DBSCAN

- A cluster is a set of core samples that can be built by recursively
  - taking a core sample, finding all of its neighbours that are core samples
  - finding all of their neighbours that are core samples
  - and so on





Comments on DBSCAN

✅ Advantages

- ✓ No need to specify number of clusters (k)
- ✓ Finds clusters of arbitrary shapes
- ✓ Handles noise/outliers naturally
- ✓ Works well for spatial / density-based data/ large datasets
-

❌ Limitations

- ✕ Sensitive to parameter choice (eps, minPts)
- ✕ Struggles with varying densities
- ✕ Performance can degrade in high dimensions
- ✕ Distance metric choice matters